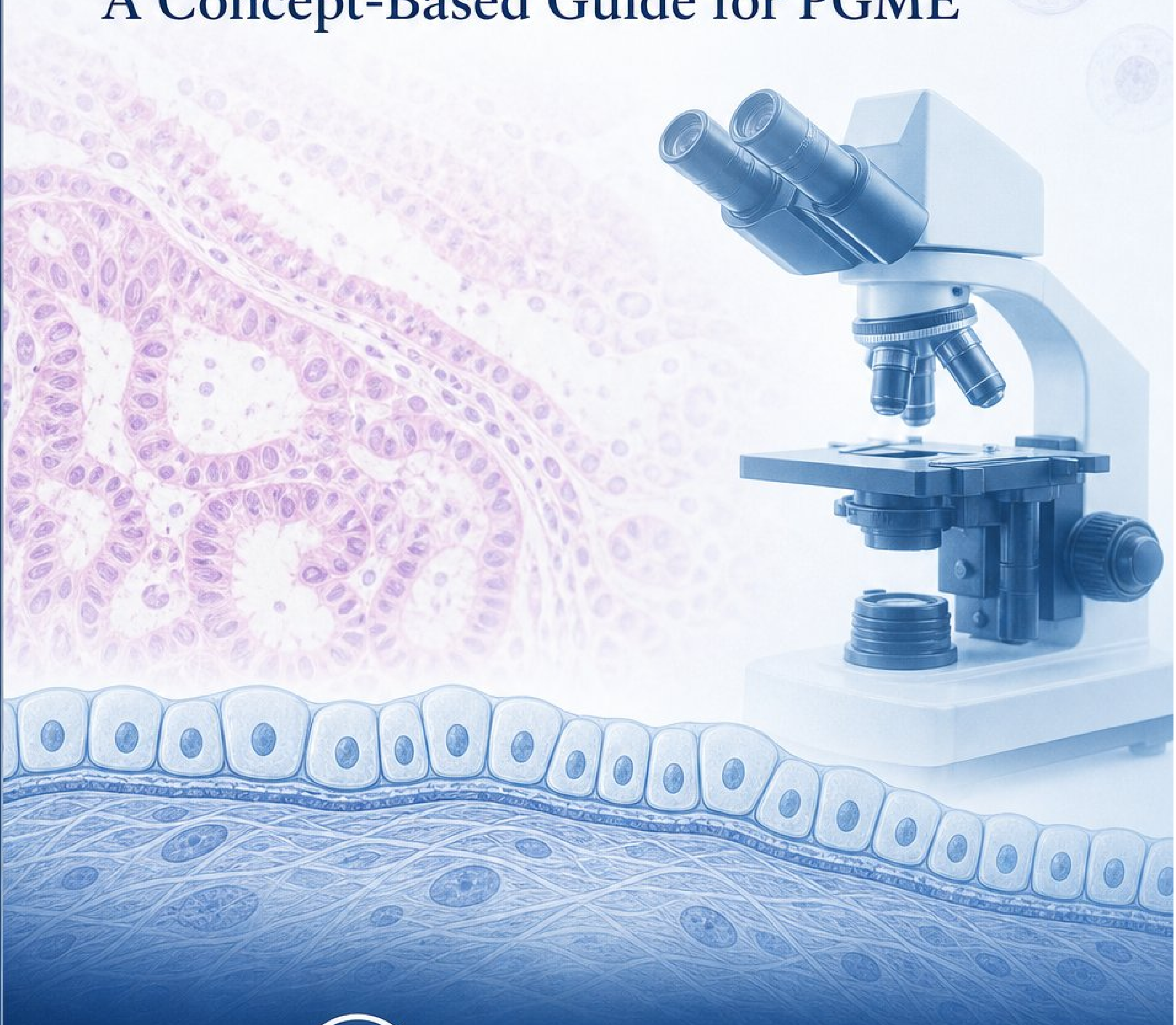
Essential Histology

A Concept-Based Guide for PGME



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A Concept-Based Guide for PGME

AREMS-HY

Primary Reference: Junqueira's Basic Histology: Text and Atlas — 17th Edition, by Anthony L. Mescher, McGraw Hill

Official Syllabus: Junqueira 17th Chapters 1,2,3,4,5,6,9,11,12,13,14,15,16,17,18,20 — 16 chapters

Reference Statement

This book is an educational synthesis aligned primarily with Junqueira's Basic Histology: Text and Atlas, 17th Edition, by Anthony L. Mescher, PhD, McGraw Hill. It does not reproduce Junqueira verbatim. It is a learning system that makes required histology understandable, memorable, connected, and teachable for the Afghanistan 1405 Clinical Medicine postgraduate examination official syllabus (16 chapters).

How to Use This Book

This book is designed as one coherent intellectual journey, not a collection of chapters. Each chapter follows a natural rhythm: Question → Why → Objectives → Landscape → Principle → Building → Seeing → Clinical → Remember → Check → Bridge. The page is intentionally calm. Emphasis is selective. Tables are for comparison, not decoration. The goal is mastery, not memorization.

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AREMS-HY — Academic Series

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This book is designed as one coherent intellectual journey, not a collection of chapters. Each chapter follows a natural rhythm:

Question → Why → Objectives → Landscape → Principle → Building → Seeing → Clinical → Remember → Check → Bridge

- Opening Question creates curiosity
- Why This Matters gives clinical relevance
- Learning Objectives tell you what you will be able to do
- The Landscape shows where you are in the journey
- The Core Principle gives you the single unifying idea
- Building the System builds from appearance to structure to composition to mechanism to function
- Seeing and Distinguishing teaches recognition and unmistakable distinctions
- Clinical Meaning shows what is clinically important
- What to Remember consolidates Must Know / Must Distinguish / Must Understand
- Check Your Understanding requires retrieval, not just recall
- Bridge connects naturally to next chapter

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Chapter 1 — Histology & Its Methods of Study

Junqueira 17th Edition — Chapter 1

> Contrast is not decoration. It is chemistry made visible — each stain and microscope reveals a specific chemical or physical property.

Opening Question

How does a pathologist turn a lump of tissue into a diagnosis you can see under a microscope — and why does the choice of stain decide what you will find?

Why This Matters

Every biopsy is a question: which molecule is accumulating, which cell is proliferating, which barrier is broken? No single technique shows everything. The fixative, the solvent, the dye, and the microscope each reveal one property and hide others. Understanding this logic prevents misdiagnosis and tells you which stain to order next.

Learning Objectives

By the end of this chapter, you will be able to:

1. Explain the physical and chemical purpose of each step in tissue processing and predict artifact when a step is skipped or delayed.
2. Derive why nucleus stains blue and cytoplasm pink in H&E from dye charge chemistry.
3. Match special stains to molecular targets (PAS, silver, trichrome, Orcein, Oil Red O, Congo red, Prussian blue, Von Kossa, Alcian blue, Feulgen) and state processing requirements.
4. Select appropriate microscopy modality for a stated investigative goal (live cells, thick specimen, surface topography, nanometer detail, membrane faces, gene copy number).
5. Explain why light resolution is limited to ~0.2 µm and why electron microscopy exceeds it.
6. Interpret staining patterns (basophilia, acidophilia, metachromasia, birefringence, empty vacuoles) as molecular content.

The Landscape

A tissue is transparent and fragile. Histology solves two problems:

1. Make it hard enough to cut — fixation → dehydration → clearing → embedding
2. Create contrast — charge (H&E), oxidation (PAS, Feulgen), metal binding (silver), solubility (lipid dyes), light interaction (birefringence, fluorescence, electron scatter)

No method is universal. Each technique = one type of contrast. If a structure is absent, ask whether the method removed it.

The Core Principle

Each technique reveals primarily the specific property it was designed to detect; no method is universal. The result must be interpreted as chemistry, not as magic color.

> CORE IDEA: Contrast is not decoration. It is chemistry made visible — each stain and microscope reveals a specific chemical or physical property.

Building the System

Living tissue autolyzes. Enzymes digest the tissue within minutes after blood supply stops. Fixation arrests this.

Fixation: cross-links macromolecules.

- Formalin (formaldehyde): reacts with amino/imino groups → methylene bridges. Standard for light microscopy. Preserves proteins adequately, does not preserve lipids well, moderate ultrastructure.

- Glutaraldehyde: dialdehyde, faster and more extensive cross-linking. Standard primary fixative for electron microscopy.
- Osmium tetroxide: binds phospholipid head groups of membranes, fixing lipids, and as a heavy metal scatters electrons → provides contrast in TEM. It is both fixative and contrast agent.

Dehydration: water removed by ascending ethanol series (70% → 95% → 100%). Required because paraffin is hydrophobic.

Clearing: ethanol removed by xylene (or substitute). Xylene is miscible with both ethanol and paraffin, making tissue translucent (cleared) and paraffin-miscible.

Embedding: paraffin infiltrates and hardens tissue into cuttable block.

Sectioning: paraffin sections ~5 μm (range ~1-10 μm). Semi-thin resin sections for EM orientation ~1 μm stained with toluidine blue. Ultrathin EM sections ~50-90 nm.

Staining: creates contrast.

H&E — charge-based:

- Hematoxylin: basic dye, positively charged. Binds negatively charged phosphate backbone of DNA/RNA → blue/purple. Term: basophilic.
- Eosin: acidic dye, negatively charged. Binds positively charged basic side chains (lysine, arginine) of proteins and collagen → pink/red. Term: acidophilic/eosinophilic.
- Deliberate inversion that confuses students: basophilic structures are acidic (they attract basic dye).

PAS — oxidation: periodic acid oxidizes hexose glycols to aldehydes → Schiff reagent binds → magenta. PAS is primarily used to demonstrate carbohydrate-rich structures — glycogen, mucin, basement membrane glycoproteins (laminin, perlecan, entactin associated with type IV collagen network), fungal walls — and is not a collagen-specific stain. Basement membranes are PAS-positive because of their carbohydrate-rich components, not because PAS stains collagen directly.

Silver impregnation — argyrophilia: reticular fibers (type III collagen) reduce silver salts → black. Reticular fiber = type III collagen in delicate stroma (liver, lymphoid organs, endocrine glands).

Masson trichrome — differential uptake: collagen blue/green vs muscle/cytoplasm red. Fibrosis assessment.

Orcein / Verhoeff — elastic: elastic fibers (elastin core + fibrillin microfibrils) → brown-black.

Lipid dyes — solubility: Oil Red O, Sudan black. Lipid dissolves in ethanol/xylene, so routine paraffin processing removes it, leaving empty vacuoles. To demonstrate lipid, need frozen section (no solvents) + lipophilic dye → red/black droplets.

Congo red — orientation: amyloid β-sheet fibrils bind Congo red and orient dye molecules → apple-green birefringence under polarized light. Birefringence alone is not diagnostic; combination of Congo red + green birefringence is.

Prussian blue (Perls) — chemical reaction: ferric iron + potassium ferrocyanide → ferric ferrocyanide (blue precipitate). Detects hemosiderin.

Von Kossa — indirect mineral demonstration: silver nitrate reacts with phosphate and carbonate anions that are bound to calcium in mineralized deposits. Silver substitutes, then reduced to metallic silver → black. It does NOT stain calcium ion directly; it demonstrates sites of mineralization where calcium phosphate/carbonate has deposited. Use as indirect indicator of calcification.

Alcian blue — acidic mucin: sulfated and carboxylated glycosaminoglycans → blue.

Feulgen — DNA: acid hydrolysis exposes aldehydes on deoxyribose → Schiff → magenta. Specific for DNA amount.

Each stain answers one question: where is nucleic acid, protein, carbohydrate, reticular fiber, collagen vs muscle, elastic fiber, lipid, amyloid, iron, mineralization, acidic mucin, DNA?

Why does blue cytoplasm mean protein synthesis? Plasma cell cytoplasm is basophilic because it is packed with rough ER and ribosomes — RNA phosphate negative → binds hematoxylin blue. Muscle fiber is eosinophilic because it is packed with contractile proteins — basic residues positive → binds eosin pink. Staining reflects dominant macromolecule.

By mechanism:

- Charge: H&E
- Oxidation→Schiff: PAS, Feulgen
- Metal reduction: silver, Von Kossa
- Lipophilic: Oil Red O
- Birefringence: Congo red + polarizer, collagen intrinsic
- Immunologic: IHC, immunofluorescence
- Nucleic acid hybridization: FISH
- Isotope incorporation: autoradiography

Microscopy by contrast:

- Phase-contrast/DIC: refractive index → live unstained cells
- Dark-field: scattered light → bright object on black background (spirochetes)
- Fluorescence: fluorochrome excitation/emission → specific molecules, DAPI DNA

- Confocal: pinhole rejects out-of-focus light → crisp optical section of thick specimen
- Polarizing: birefringence → amyloid, collagen, crystals
- TEM: electron scatter → internal ultrastructure 2-D, ~0.1 nm resolution
- SEM: secondary electrons → 3-D surface topography
- Freeze-fracture: fracture through hydrophobic interior of lipid bilayer → E-face (outer leaflet inner aspect) and P-face (inner leaflet outer aspect) with intramembrane particles

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Seeing and Distinguishing

Recognition Logic

Look for... blue nucleus + pink cytoplasm = H&E baseline.

Then confirm with... magenta = PAS carbohydrate; black network = silver reticular; blue/green vs red = trichrome collagen vs muscle; red droplets on frozen = lipid; apple-green birefringence after Congo red under polarized light = amyloid; blue granules = iron Perls; black deposits after silver nitrate = mineralization (Von Kossa, indirect).

Do not confuse with... metachromasia (toluidine blue → purple/red on dense polyanions like heparin in mast cells due to dye polymerization) is not basophilia. Birefringence of collagen (white) is not Congo red green.

Decisive feature is... charge chemistry for H&E, oxidation for PAS, silver reduction for reticular and mineralization, solubility for lipid, polarization for amyloid.

Compare & Distinguish

Clinical Meaning

Feature	Frozen Section	Paraffin Section	
Time	~minutes	hours–days	
Lipid	Preserved	Dissolved (empty vacuoles)	
Enzyme activity	Preserved	Lost	
Morphology	Inferior, ice crystal artifacts	Superior, best detail	
Use	Intraoperative diagnosis, lipid, enzymes	Routine diagnosis, long storage	
Feature	TEM	SEM	Light
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Feature	Frozen Section	Paraffin Section	
Signal	Transmitted electrons, electron scatter	Secondary electrons from surface	Visible light absorption/transmission
Image	2-D internal	3-D surface	2-D color
Resolution	~0.1 nm	~1-10 nm	~0.2 µm
Best for	Organelles, filtration barrier, foot processes	Cilia, microvilli surface	Survey, H&E

Fatty liver vs glycogen storage vs hemochromatosis vs amyloidosis: same hepatomegaly, different stain. Fat needs frozen + Oil Red O because routine processing dissolves lipid → empty vacuoles on H&E. Glycogen needs PAS (magenta) and diastase sensitivity. Iron needs Prussian blue. Amyloid needs Congo red + polarized apple-green. Von Kossa would show black only if calcification present, and indicates phosphate/carbonate of mineral, not calcium ion itself.

Frozen section: surgeon needs margin in 15 minutes. Paraffin cannot be done that fast. Frozen short-circuits dehydration/clearing/embedding, preserves enzymes/lipids, but ice crystals create artifacts.

Goodpasture/Alport: basement membrane best shown by PAS (glycoproteins type IV collagen + laminin). Thickened in diabetes.

What to Remember

High-Yield Knowledge

Must Know:

- Basophilic = nucleic acid = binds basic dye hematoxylin = blue. Acidophilic = protein = binds acidic dye eosin = pink.
- PAS = carbohydrate (glycogen, mucin, basement membrane, fungi) — not collagen, not reticular.
- Reticular fibers = type III collagen = silver black.
- Collagen vs muscle = Masson trichrome.
- Elastic fibers = Orcein/Verhoeff.
- Lipid = frozen + Oil Red O (routine processing dissolves it).
- Amyloid = Congo red + apple-green birefringence polarized.
- Iron = Prussian blue. Mineralization = Von Kossa (phosphate/carbonate, silver black, indirect indicator of calcium).
- LM fixative = formalin. EM = glutaraldehyde + osmium (osmium = membranes + contrast).
- Light resolution ~0.2 µm, EM ~0.1 nm (shorter wavelength).

Must Distinguish:

- Basophilic vs metachromatic (blue vs purple shift due to polymerization)
- Silver reticular (type III) vs trichrome collagen (type I) vs Orcein elastic
- TEM inside vs SEM surface vs freeze-fracture membrane faces
- Frozen (lipid, enzymes, speed, inferior morphology) vs paraffin (best morphology, lipid lost)

Must Understand:

- Why empty vacuole on H&E does not mean empty cell — lipid dissolved.
- Why Von Kossa is indirect.
- Why resolution ≠ magnification.

Common Misconceptions

Misconception: "Clearing removes water."

Reality: Dehydration (ethanol) removes water. Clearing (xylene) removes ethanol and makes tissue miscible with paraffin. Each step exists because no single reagent bridges water and paraffin.

Misconception: "Von Kossa stains calcium."

Reality: Von Kossa demonstrates phosphate and carbonate anions in mineralized matrix via silver substitution. Calcium is inferred because calcium phosphate/carbonate is the mineral, but the reaction detects anion, not calcium ion. Always report as "mineralization" or "calcium phosphate deposits, Von Kossa positive (indirect)".

Misconception: "More magnification = more resolution."

Reality: Resolution = minimum distance two points can be separated as distinct, set by wavelength and numerical aperture. Light limited to ~0.2 µm by visible light wavelength (~400-700 nm). Electron wavelength much shorter → ~0.1 nm. Magnification without resolution is empty magnification.

Misconception: "H&E shows everything."

Reality: H&E shows charge distribution only. Lipid, glycogen amount, reticular fibers, elastic fibers, iron, mineral, amyloid need special stains.

Integrated Summary

Histology makes transparent fragile tissue cuttable and contrasted. Processing is miscibility ladder: water→alcohol→xylene→paraffin, each step bridging to next; fixation cross-links to prevent autolysis. Staining is chemistry: charge, oxidation, metal reduction, solubility, birefringence, antibody-antigen, probe hybridization, isotope incorporation. Microscopy is optics: refractive index, fluorescence, birefringence, electron scatter. Choose fixative, stain, microscope to answer specific molecular question. No technique universal.

Check Your Understanding

1. Explain why dehydration and clearing are two separate steps, not one.

2. A plasma cell is intensely basophilic while skeletal muscle is eosinophilic — explain from dominant macromolecule and dye chemistry.

3. A researcher wants to prove active DNA synthesis vs amount of DNA vs cycling marker — compare ^{3}H-thymidine autoradiography, Feulgen, Ki-67 IHC.

4. Why is TEM required to see podocyte foot process effacement in minimal change disease?

5. How do you distinguish amyloid birefringence from collagen birefringence?

6. Why does freeze-fracture split membrane into two faces rather than separating membrane from cytoplasm?

7. A signet-ring cell with single large clear space and flattened nucleus on H&E — most safe interpretation and how to confirm?

Bridge

You have now seen how tissue is prepared and stained. Next, step inside the cell itself — the machinery that every stain in this chapter is lighting up: the cytoplasm.

Rapid Review

- Processing: fixation (formalin LM, glutaraldehyde+osmium EM) → dehydration (ethanol removes water) → clearing (xylene removes ethanol) → embedding (paraffin) → ~5 μm section.

- H&E: hematoxylin basic + binds DNA/RNA phosphate → blue basophilic; eosin acidic − binds protein basic residues → pink acidophilic.

- PAS magenta = carbohydrate. Silver black = reticular type III. Trichrome blue/green collagen vs red muscle. Orcein = elastic. Oil Red O red = lipid, needs frozen. Congo red + polarized green = amyloid. Perls blue = iron. Von Kossa black = mineral phosphate/carbonate indirect.

- Microscopes: phase live, confocal pinhole thick, TEM internal 0.1 nm, SEM surface, freeze-fracture E/P faces.

- Frozen = speed + lipid + enzymes, inferior morphology. Paraffin = best morphology, lipid lost.

Chapter 2 — The Cytoplasm

Junqueira 17th Edition — Chapter 2

> The cytoplasm organizes work by compartment: RER synthesizes proteins for export, SER synthesizes lipids and handles detoxification and calcium, Golgi modifies and sorts, lysosomes digest, peroxisomes oxidize, mitochondria produce ATP, and the cytoskeleton organizes space and transport.

Opening Question

How does a cell build, move, and ship thousands of proteins without mixing them up?

Why This Matters

Pancreatic acinar cells secrete amylase, plasma cells secrete antibody, liver cells detoxify drugs — same basic machinery, different outputs. Kartagener syndrome (immotile cilia), I-cell disease (lysosomal enzymes misrouted), and microvillus inclusion disease all trace to cytoplasmic machinery failure.

Learning Objectives

By the end of this chapter, you will be able to:

1. Describe membrane flow from RER → Golgi → vesicles and explain how signal sequences and M6P targeting determine destination.
2. Distinguish RER, SER, Golgi, lysosome, peroxisome, mitochondria by structure, function, and microscopic appearance.
3. Classify cytoskeletal filaments by diameter, protein, stability, and function (microfilaments actin 6-8 nm, intermediate 10 nm, microtubules 25 nm).
4. Explain why basophilic cytoplasm indicates RER-rich protein synthesis.
5. Predict consequence of defects in microtubule motors, dynein arms, lysosomal targeting, peroxisomal oxidases.

The Landscape

Cytoplasm = cytosol (aqueous phase) + organelles (membrane-bound compartments) + cytoskeleton (scaffolding and tracks) + inclusions (stored products).

Governing principle: compartmentalization by membrane + transport by vesicles + positioning by cytoskeleton.

The Core Principle

The cell solves crowding by membranes that create distinct chemical environments, and solves movement by cytoskeletal tracks and motors.

CORE IDEA: The cytoplasm organizes work by compartment: RER synthesizes proteins for export, SER synthesizes lipids and handles detoxification and calcium, Golgi modifies and sorts, lysosomes digest, peroxisomes oxidize, mitochondria produce ATP, and the cytoskeleton organizes space and transport.

Building the System

RER: studded with ribosomes (rRNA basophilic). Co-translational insertion: signal sequence → SRP → translocon → lumen. Protein synthesis for secretion, lysosomal enzymes, membrane proteins. Extensive in protein-secreting cells (pancreas, plasma cell) → basophilic cytoplasm.

SER: no ribosomes. Lipid synthesis, steroid synthesis (adrenal cortex fasciculata, Leydig), detoxification (smooth ER proliferation with drugs, cytochrome P450), calcium storage (muscle sarcoplasmic reticulum).

Golgi: cis (receiving, near RER) → medial → trans (shipping) + TGN (trans-Golgi network). Glycosylation, sulfation, sorting: M6P for lysosomal enzymes, signal peptides for secretion. Polarized: cis convex, trans concave with vesicles.

Lysosome: acid hydrolases (acid phosphatase marker) active at ~pH 5, proton pump. Primary (enzyme only) + secondary (enzyme + substrate). M6P receptor targets enzymes to lysosome; defect → I-cell disease enzymes secreted extracellularly, inclusions.

Peroxisome: oxidase + catalase. Beta-oxidation of very long chain fatty acids, detox H_2O_2, not acid hydrolases. Distinct from lysosome. Marker: catalase. Defect → Zellweger.

Mitochondria: double membrane, outer smooth, inner cristae (tubular in steroid cells), matrix with enzymes, mtDNA, ribosomes. ATP synthesis, apoptosis, calcium. Eosinophilic cytoplasm when abundant.

Cytoskeleton:

- Microfilaments: ~6-8 nm, actin + myosin, dynamic, cortex, microvilli core, contractile ring, focal adhesions. Polarity barbed/pointed, ATP.
- Intermediate filaments: ~10 nm, stable, cell-type specific: keratin (epithelium), vimentin (mesenchyme), desmin (muscle), neurofilament (neuron), GFAP (astrocyte), lamins (nuclear). Anchors desmosomes and hemidesmosomes.
- Microtubules: ~25 nm, tubulin α/β dimer, dynamic instability, GTP, MTOC centrosome, 9+2 in cilia, tracks for kinesin (anterograde) and dynein (retrograde). Mitotic spindle.

Why does pancreatic acinar cell have basal basophilia and apical eosinophilia? Basal RER (basophilic) synthesizes zymogen, apical zymogen granules (eosinophilic protein) store enzymes. Polarity reflects flow.

Why does microvillus have actin core while cilium has microtubule core? Microvillus needs stiff stable scaffold to increase surface area without movement → actin. Cilium needs motility → microtubule doublet sliding via dynein.

Organelles by membrane: single membrane (RER, SER, Golgi, lysosome, peroxisome) vs double (mitochondria, nucleus). Filaments by size and protein.

Seeing and Distinguishing

Recognition Logic

Look for... basal blue cytoplasm + apical pink granules = protein-secreting cell (RER → zymogen). Central dark nucleus + basophilic cytoplasm + clock-face chromatin = plasma cell (RER-rich antibody).

Then confirm with... microvilli (actin, no motility, brush border) vs cilia (9+2 microtubule, motile) vs stereocilia (long branched microvilli, not true cilia, epididymis). Terminal web (actin) under microvilli.

Do not confuse with... RER basophilia (ribosomes) vs heterochromatin basophilia (DNA). Both blue but location and cell type differ.

Decisive feature: ~6-8 nm actin vs ~10 nm intermediate vs ~25 nm tubulin; 9+2 microtubule for motile cilia; M6P for lysosome targeting.

Compare & Distinguish

Clinical Meaning

Feature	RER	SER		
Ribosomes	Present	Absent		
Function	Protein for export/membrane/lysosome	Lipid synthesis, detox, Ca2+ storage		
Stain	Basophilic	Not basophilic		
Cell example	Plasma cell, pancreatic acinar	Hepatocyte, adrenal fasciculata, muscle		
Feature	Lysosome	Peroxisome		
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Enzymes	Acid hydrolases, acid phosphatase	Oxidase, catalase		
pH	Acidic ~5	Not acidic		
Marker	Acid phosphatase, M6P	Catalase		
Origin	Golgi	ER budding, self-replication		

Feature	RER	SER		
Defect	I-cell (M6P)	Zellweger (biogenesis)		
Filament	Diameter	Protein	Function	Cell-type marker
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Microfilament	~6-8 nm	Actin	Cortex, microvilli, motility	—
Intermediate	~10 nm	Keratin, vimentin, desmin, neurofilament, GFAP	Mechanical stability, desmosome anchor	Epithelial vs mesenchymal vs muscle vs neuron
Microtubule	~25 nm	Tubulin	Transport, cilia, mitosis	—

Kartagener syndrome: dynein arm defect → immotile cilia → bronchiectasis, sinusitis, situs inversus (nodal cilia defect). Histology: cilia present but not motile, EM shows missing dynein arms.

I-cell disease (mucolipidosis II): GlcNAc phosphotransferase defect → no M6P tag → lysosomal enzymes not targeted to lysosome → secreted, lysosomes lack enzymes → inclusion bodies (phase-dense inclusions).

Microvillus inclusion disease: actin cytoskeleton defect → atrophic microvilli.

What to Remember

High-Yield

Must Know: RER basophilic protein synthesis, SER lipid/detox/Ca2+, Golgi cis→trans glycosylation + sorting, lysosome acid hydrolase M6P, peroxisome oxidase/catalase, mitochondria double membrane cristae. Filament diameters ~6-8, ~10, ~25 nm. Actin microvilli, tubulin cilia 9+2.

Must Distinguish: RER vs SER, lysosome vs peroxisome, microvilli actin vs cilia tubulin vs stereocilia long microvilli.

Must Understand: Membrane flow polarity, why basophilic cytoplasm means synthetic activity, why cytoskeleton cell-type specific intermediate filaments are diagnostic in pathology.

Common Misconceptions

Misconception: "SER is only for detox."

Reality: SER also synthesizes lipid, steroid, stores Ca2+. In muscle, SER is sarcoplasmic reticulum Ca2+ release for contraction.

Misconception: "Lysosome and peroxisome are similar because both digest."

Reality: Lysosome digests via acid hydrolases, acidic pH, M6P targeted from Golgi. Peroxisome oxidizes via oxidase/catalase, neutral, ER-derived, handles very long chain fatty acids and H_2O_2.

Misconception: "All basophilia is nucleus."

Reality: Cytoplasmic basophilia = RER/ribosomes = protein synthesis.

Integrated Summary

Cytoplasm organizes chemistry by membranes: RER makes proteins for export/membrane/lysosome, SER makes lipid and handles calcium/detox, Golgi modifies and sorts via M6P and signal peptides, lysosome digests at acidic pH, peroxisome oxidizes via catalase, mitochondria makes ATP. Cytoskeleton provides scaffold and tracks: actin for cortex and microvilli, intermediate filaments for mechanical stability and cell identity, microtubules for transport and cilia. Structure predicts staining and function.

Check Your Understanding

1. Explain why plasma cell cytoplasm is intensely basophilic and where antibody goes after synthesis.
2. Reconstruct membrane flow from signal sequence to secreted protein, including M6P branch to lysosome.
3. Compare microvillus, cilium, stereocilium by core protein, motility, location.
4. Predict consequence of kinesin vs dynein motor defect.
5. Why does I-cell disease cause extracellular lysosomal enzymes and intracellular inclusions?

Bridge

The cytoplasm builds and moves. The nucleus controls what is built, when, and how much. Next, how two meters of DNA fits in a six micrometer nucleus and remains readable.

Rapid Review

- RER basophilic = protein for export/membrane/lysosome. SER = lipid, steroid, detox, Ca^{2+}.
- Golgi cis→trans, glycosylation, sorting, M6P lysosome.
- Lysosome acid hydrolase pH5, peroxisome oxidase/catalase.
- Filaments: actin ~6-8 nm microvilli, intermediate ~10 nm keratin/vimentin/desmin/neurofilament/GFAP, microtubule ~25 nm tubulin 9+2 cilia.
- Kartagener dynein immotile cilia, I-cell M6P defect.

Chapter 3 — The Nucleus

Junqueira 17th Edition — Chapter 3

> A pale, euchromatic nucleus with a prominent nucleolus indicates active transcription and ribosome production, while a dark, heterochromatic nucleus indicates quiescence.

Opening Question

How does 2 meters of DNA fit in a 6 μm nucleus and still remain readable?

Why This Matters

Cancer diagnosis rests on nuclear atypia: irregular envelope, coarse chromatin, prominent nucleoli. Barr body (inactive X) explains dosage compensation. Understanding euchromatin vs heterochromatin explains why some cells transcribe actively and others do not.

Learning Objectives

By the end of this chapter, you will be able to:

1. Describe nuclear envelope (double membrane, perinuclear space, pores ~120 MDa, lamina) and its functions.
2. Distinguish euchromatin (active, light, dispersed) vs heterochromatin (inactive, dark, condensed) functionally and microscopically.
3. Explain nucleolus organization (FCC, DFC, GC) and its role in rRNA transcription and ribosome biogenesis.
4. Predict consequence of lamin defects and nuclear pore transport defects.
5. Recognize nuclear features of protein-synthesizing vs inactive cells.

The Landscape

Nucleus = genome storage + reading control. Envelope = barrier + transport. Chromatin = DNA + histones, organization around accessibility. Nucleolus = ribosome factory.

The Core Principle

Chromatin organization controls access to genetic information. Euchromatin open for transcription, heterochromatin closed.

> CORE IDEA: A pale, euchromatic nucleus with a prominent nucleolus indicates active transcription and ribosome production, while a dark, heterochromatic nucleus indicates quiescence.

Building the System

Nuclear envelope: double membrane, outer continuous with RER, perinuclear space, inner with lamina (lamin intermediate filaments) for mechanical support. Pores: large ~120 MDa complexes, selective transport, importin/exportin, Ran-GTP.

Chromatin: DNA + histone octamer → nucleosome → 11 nm fiber → 30 nm fiber → looped domains. Euchromatin: dispersed, transcriptionally active, light in LM, peripheral decondensed. Heterochromatin: condensed, inactive, dark, often peripheral under envelope and around nucleolus. Constitutive (always condensed, centromere) vs facultative (inactive X Barr body).

Chromosome: condensed chromatin during mitosis, centromere, telomere.

Nucleolus: not membrane-bound, organizer regions (NOR) contain rDNA. Zones: fibrillar center (rDNA), dense fibrillar component (transcription), granular component (processing). Prominent in protein-synthesizing cells (plasma cell, neuron). Multiple nucleoli in growing cells.

Why does active cell have large pale nucleus with prominent nucleolus? Active transcription needs dispersed euchromatin (light) and ribosome biogenesis (nucleolus prominent). Inactive lymphocyte has small dark heterochromatic nucleus with no nucleolus — minimal transcription.

Why does nuclear envelope have pores? Nucleus needs to export mRNA and ribosomal subunits and import proteins and transcription factors, but keep DNA separate. Pores provide selective gated transport.

Chromatin: euchromatin vs heterochromatin (constitutive vs facultative). Nucleolus: 1-2 prominent in active cells, inconspicuous in inactive.

Seeing and Distinguishing

Recognition Logic

Look for... pale nucleus + prominent nucleolus = active protein synthesis (plasma cell, neuron, hepatocyte). Dark small nucleus = inactive (lymphocyte, sperm).

Then confirm with... Barr body at periphery in female somatic cells = inactive X, heterochromatin.

Do not confuse with... nucleolus vs inclusion: nucleolus inside nucleus, inclusion in cytoplasm.

Decisive feature: euchromatin light central, heterochromatin dark peripheral, nucleolus prominent when RER-rich cytoplasm basophilic — coupled.

Compare & Distinguish

Clinical Meaning

Feature	Euchromatin	Heterochromatin
Stain	Light, pale	Dark, basophilic
Activity	Active transcription	Inactive
Location	Central, dispersed	Peripheral, around nucleolus, Barr body

Feature	Euchromatin	Heterochromatin
Example	Plasma cell euchromatic, neuron	Small lymphocyte, inactive
Feature	Nuclear envelope	Plasma membrane
Layers	Double, perinuclear space	Single bilayer
Pores	Large 120 MDa, selective	No pores, channels/transporters
Lamina	Lamin intermediate filaments	No lamina, cortex actin

Cancer atypia: irregular nuclear envelope, coarse clumped chromatin, multiple prominent nucleoli, high N:C ratio. Reflects abnormal transcription and proliferation.

Laminopathies: lamin A/C mutations → muscular dystrophy, progeria — nuclear fragility.

Barr body: buccal smear, sex chromatin, dosage compensation.

What to Remember

High-Yield

Must Know: Envelope double + pores 120 MDa + lamina lamin; euchromatin light active, heterochromatin dark inactive; nucleolus rRNA factory prominent in protein synthesis; Barr body inactive X peripheral.

Must Distinguish: Euchromatin vs heterochromatin, nucleolus vs inclusion.

Must Understand: Why active cells have pale nuclei with prominent nucleoli and basophilic cytoplasm — coupled transcription and translation demand.

Common Misconceptions

Misconception: "Heterochromatin is junk."

Reality: Heterochromatin is inactive but functional: centromere stability, dosage compensation (Barr body), gene silencing.

Misconception: "Nucleolus is membrane-bound organelle."

Reality: Nucleolus is non-membrane compartment formed around NORs, dynamic.

Integrated Summary

Nucleus stores genome in chromatin whose condensation controls access: euchromatin open and light for active transcription, heterochromatin closed and dark for silencing. Envelope double membrane with selective pores and lamin support separates but connects to cytoplasm. Nucleolus around rDNA makes rRNA and ribosomal subunits, prominent when protein synthesis high. Nuclear morphology predicts cell activity.

Check Your Understanding

1. Explain why small lymphocyte nucleus is dark and plasma cell nucleus is pale with prominent nucleolus.

2. Describe nuclear pore function and why it needs to be large and selective.

3. What is Barr body and why at periphery?

4. How does lamin defect cause disease?

Bridge

Cells with controlled genomes organize into tissues. Next, how polarized cells form sheets that seal, absorb, and secrete: epithelial tissue.

Rapid Review

- Envelope double + perinuclear space + pores 120 MDa + lamina lamin.

- Euchromatin light active central, heterochromatin dark inactive peripheral/Barr.

- Nucleolus FCC/DFC/GC, rRNA, prominent in active cells.

- Active cell: pale nucleus + prominent nucleolus + basophilic cytoplasm.

Chapter 4 — Epithelial Tissue

Junqueira 17th Edition — Chapter 4

> Epithelial polarity solves three different problems: the apical surface handles the lumen, the lateral surface handles neighboring cells, and the basal surface handles attachment to the basement membrane.

Opening Question

Why does acid reflux slowly turn the lining of the food pipe into intestine — and why does that raise the risk of cancer?

Why This Matters

Esophagus is stratified squamous for abrasion resistance; intestine is columnar with goblet cells for absorption and lubrication. Chronic acid stress reprograms stem cells → metaplasia: Barrett esophagus, premalignant. Recognizing epithelium type and metaplasia is daily pathology.

Learning Objectives

By the end of this chapter, you will be able to:

1. State defining features of epithelium (polarity, avascularity, basement membrane, regeneration) and explain functional necessity.
2. Classify epithelium by layers and shape and by special types (pseudostratified, transitional) with locations.
3. Explain junctional complex order apical→basal and molecule + cytoskeletal anchor for each junction, and predict blister level when disrupted.
4. Describe basement membrane composition (type IV collagen network, laminin, entactin/nidogen, perlecan) and distinguish basal lamina vs reticular lamina.
5. Distinguish microvilli (actin), cilia (9+2 tubulin), stereocilia (long branched microvilli), primary cilium (9+0 sensory).
6. Classify glands by release (merocrine/apocrine/holocrine), secretion (serous/mucous), duct architecture, and myoepithelial role.

The Landscape

Epithelium is boundary tissue. An epithelial cell is polarized: apical (lumen) specialized for protection/absorption/transport, lateral (neighbors) for adhesion/sealing/communication, basal (matrix) for anchorage/filtration. Entire epithelium organized so nothing crosses except through cells.

The Core Principle

Polarity + avascularity + basement membrane + regeneration = epithelium. Form follows function: flat for exchange, tall for absorption, layered for protection, dome-shaped for stretch.

CORE IDEA: Epithelial polarity solves three different problems: the apical surface handles the lumen, the lateral surface handles neighboring cells, and the basal surface handles attachment to the basement membrane.

Building the System

Defining features: Polarity (different proteins apical/lateral/basal), Avascular (no vessels, diffusion across BM limits thickness), Basement membrane (LM term) = basal lamina (epithelial: lamina lucida + lamina densa) + reticular lamina (fibroblast type III). Molecular: type IV collagen sheet-forming network (not fibrillar, mesh via NC1) + laminin + entactin/nidogen + perlecan. Regeneration via basal stem cells.

Classification: Layers simple vs stratified vs pseudostratified (all touch BM nuclei different heights, simple) vs transitional (umbrella dome). Shape squamous/cuboidal/columnar. Name by surface cells.

Junctional complex — order logical: most apical must seal first.

- Tight (zonula occludens): most apical, seal, claudin/occludin/JAMs, ZO-1 actin, barrier/fence.
- Adherens (zonula adherens): E-cadherin + catenins + vinculin + actin.
- Desmosome (macula adherens): desmoglein/desmocollin + plakoglobin + desmoplakin + keratin, intercellular bridges spines.
- Gap: connexin communication.
- Hemidesmosome basal: integrin α6β4 binds laminin-332 (not directly type IV), BP180/230 + plectin + keratin, anchors to BM. Laminin binds type IV network.

Apical specializations: Microvilli actin core 20-30 filaments + villin fimbrin terminal web actin increase area 20-30x non-motile. Cilia 9+2 microtubule + dynein motile basal body. Stereocilia long branched microvilli actin non-motile epididymis inner ear. Primary cilium 9+0 sensory.

Glands: Release merocrine exocytosis no loss, apocrine apical loss historically mammary now merocrine, holocrine whole cell sebaceous. Secretion serous protein basophilic basal eosinophilic apical round nucleus vs mucous pale foamy flat basal. Duct simple vs compound tubular vs acinar. Myoepithelial branched actin/myosin squeeze.

Why esophagus stratified squamous while intestine simple columnar with microvilli? Esophagus abrasion → layers protect. Intestine absorption → single tall + microvilli area without thickening.

Why tight junction most apical? Paracellular space begins at apex; seal must be apical to define outside vs inside.

Seeing and Distinguishing

Recognition Logic

Look for... sheet + BM + no vessels = epithelium. Count nuclei stacks 1=simple >1=stratified, shape surface cells, umbrella dome=transitional, all touch BM + cilia + goblet=pseudostratified, brush border=microvilli.

Then confirm with... tight claudin seal, adherens cadherin actin, desmosome desmoglein keratin, gap connexin, hemidesmosome integrin laminin.

Do not confuse with... pseudostratified is simple, stratified named by surface not basal, transitional umbrella not squamous.

Decisive: umbrella=transitional, cilia+all touch BM=pseudostratified, flat nucleated=stratified non-kerat, flat anucleate=kerat.

Compare & Distinguish

Clinical Meaning

Feature	Pseudostratified	Stratified	Transitional
All touch BM?	Yes	No	No
Special cell	Cilia+goblet or stereocilia	Flat surface	Umbrella dome
Key	All reach base	Surface shape names it	Stretchable dome
Feature	Microvilli	Cilia	Stereocilia
---	---	---	---
Core	Actin	9+2 tubulin + dynein	Actin long
Motile	No	Yes	No
Function	Area	Move fluid	Absorption/sensory

Barrett: chronic GERD → stem reprogramming squamous→columnar goblet premalignant adenocarcinoma risk, need goblet to diagnose.

Blistering: pemphigus anti-desmoglein desmosome cell-cell → intraepidermal suprabasal, pemphigoid anti-hemidesmosome BP180/230 cell-matrix → subepidermal. Level equals molecule.

What to Remember

High-Yield

Must Know: Polarity, avascular, BM type IV network + laminin perlecan entactin, basal lamina + reticular lamina III. Junctions tight claudin ZO-1 actin → adherens E-cadherin catenin actin → desmosome desmoglein desmoplakin keratin → gap connexin → hemidesmosome integrin α6β4 laminin-332 keratin basal. Microvilli actin, cilia 9+2 dynein, stereocilia long microvilli.

Must Distinguish: Simple vs pseudostratified vs stratified vs transitional, microvilli vs cilia vs stereocilia, desmosome vs hemidesmosome.

Must Understand: Thickness limited diffusion, junction order logical, shape follows function, metaplasia trades function for survival premalignant.

Common Misconceptions

Misconception: Pseudostratified is stratified. Reality: all touch basal lamina simple.

Misconception: Desmosomes use integrin. Reality: desmosomal cadherins desmoglein keratin, hemidesmosomes integrin laminin.

Misconception: Epithelium vascular. Reality: avascular diffusion-limited.

Integrated Summary

Epithelium polarized sheet on BM avascular regenerative. Apical microvilli actin absorption cilia tubulin transport, lateral junctions fixed order seal tight adherens desmosome gap, basal hemidesmosome integrin laminin keratin anchoring type IV network. Classification layers+surface shape plus special pseudostratified transitional. Glands merocrine/apocrine/holocrine serous vs mucous myoepithelial squeeze. Metaplasia stem reprogramming chronic stress premalignant.

Check Your Understanding

1. Why epithelium cannot be arbitrarily thick?
2. Reconstruct junction order with molecule anchor disease.
3. Distinguish pseudostratified vs stratified vs transitional decisive features.
4. Explain Barrett mechanism why goblet required.
5. Why pemphigus intraepidermal vs pemphigoid subepidermal?

Bridge

Epithelium is the boundary tissue. Next, the framework that supports, connects, and defends: connective tissue.

Rapid Review

- Features polarity avascular BM regeneration.
- BM type IV network + laminin entactin perlecan, basal lamina epithelial + reticular lamina III fibroblast.
- Junctions apical→basal tight claudin ZO-1 actin, adherens E-cadherin catenin actin, desmosome desmoglein desmoplakin keratin, gap connexin, hemidesmosome integrin laminin keratin basal.
- Apical microvilli actin brush, cilia 9+2 dynein, stereocilia long microvilli.
- Classification layers+surface, pseudostratified all touch BM, transitional umbrella.
- Glands merocrine holocrine serous basophilic basal eosinophilic apical vs mucous pale, myoepithelial.

- Metaplasia stem reprogramming Barrett premalignant.

Chapter 5 — Connective Tissue

Junqueira 17th Edition — Chapter 5

> In connective tissue, collagen fibers resist tension, elastic fibers provide recoil, and glycosaminoglycans hold water to resist compression.

Opening Question

Why do some people have stretchy skin and others fragile bones — and how can a single tissue explain both?

Why This Matters

Marfan (fibrillin), Ehlers-Danlos (collagen), osteogenesis imperfecta (type I collagen), scurvy (vitamin C hydroxylation) — all connective tissue diseases. Wound healing switches type III → type I collagen. Understanding fiber composition predicts disease and scar outcome.

Learning Objectives

By the end of this chapter, you will be able to:

1. State three components of connective tissue (cells, fibers, ground substance) and explain how ratio defines loose vs dense.
2. Trace collagen biosynthesis (procollagen → hydroxylation vit C → triple helix → fibril → cross-link) and explain scurvy.
3. Match collagen types I-IV to location and disease and explain fibrillar vs sheet-forming.
4. Describe elastic fiber composition (elastin core + fibrillin microfibrils) and Marfan mechanism.
5. Identify resident and transient cells by function and appearance.
6. Classify adult connective tissue (loose, dense regular, dense irregular, elastic, reticular, mucoid).

The Landscape

Connective tissue = cells + extracellular matrix; matrix is load-bearing, space-filling, signal-carrying. Molecular composition determines mechanical behavior and disease.

The Core Principle

The molecular composition of the extracellular matrix determines mechanical properties, tissue distribution, and disease vulnerability.

> CORE IDEA: In connective tissue, collagen fibers resist tension, elastic fibers provide recoil, and glycosaminoglycans hold water to resist compression.

Building the System

Three components: cells, fibers, ground substance (GAGs + proteoglycans + glycoproteins fibronectin, laminin).

Fibers:

- Collagen: most abundant protein. Biosynthesis: procollagen α chains RER → hydroxylation proline/lysine (vitamin C cofactor Fe^{2+} O_2) → triple helix → procollagen peptidase → tropocollagen → fibrils quarter-stagger cross-links lysyl oxidase → fibers. Defect hydroxylation → weak collagen → scurvy bleeding gums.
- Collagen types master table (Junqueira 17th Chapter 5):
- Type I: fibrillar thick, bone, skin, tendon, dentin, organ capsules, cornea, tensile strength, disease OI, EDS arthrochalasia.
- Type II: fibrillar thin, hyaline cartilage matrix, vitreous, nucleus pulposus.
- Type III: fibrillar thin reticular, granulation tissue, around vessels, lymphoid stroma, skin, uterus, delicate stroma, wound healing early, vascular EDS.
- Type IV: sheet-forming network via NC1 domains, basal lamina, lens capsule, filtration, Goodpasture, Alport.
- Elastic: elastin core hydrophobic desmosine/isodesmosine + fibrillin microfibrils scaffold. Development fibrillin scaffold first then elastin peripherally → mature elastin core surrounded microfibrils. Recoil Orcein/Verhoeff brown-black. Marfan fibrillin-1 defect → weak recoil → aortic aneurysm tall arachnodactyly.
- Reticular: type III collagen silver argyrophilic black network delicate stroma liver lymphoid endocrine.

Ground substance: GAGs highly hydrated hyaluronan chondroitin dermatan keratan heparan + proteoglycans aggrecan → compression water. Glycoproteins fibronectin cell adhesion laminin BM.

Cells:

- Fibroblast builder spindle euchromatic basophilic RER collagen + ground substance. Fibrocyte quiescent.
- Macrophage defense monocyte-derived kidney nucleus lysosomes phagocytosis.
- Mast allergy round metachromatic granules heparin sulfated GAG dye polymerization purple/red toluidine blue histamine heparin.
- Plasma antibody clock-face nucleus basophilic cytoplasm perinuclear halo Golgi Russell bodies.
- Adipocyte energy (see next chapter).
- Leukocytes transient.

Adult CT types (Junqueira Ch5):

- Loose (areolar): all components papillary dermis around vessels.

- Dense regular: parallel collagen type I tendon ligament fibroblasts rows.
- Dense irregular: interwoven collagen reticular dermis capsules.
- Elastic: parallel elastic fibers ligamentum nuchae large artery media.
- Reticular: type III network lymphoid.
- Mucoid: Wharton jelly umbilical cord hyaluronan-rich.

Why tendon tension one direction while dermis many? Tendon dense regular parallel type I along axis maximal one direction. Dermis dense irregular interwoven many directions.

Why GAGs important? GAGs sulfated negative hold water hydrated gel resists compression, collagen resists tension.

Seeing and Distinguishing

Recognition Logic

Look for... wavy pink collagen bundles dense CT, thin dark Orcein elastic, black silver network reticular III, metachromatic purple toluidine mast heparin, clock-face basophilic plasma.

Do not confuse with... reticular type III vs type I collagen both collagen but reticular delicate silver black I thick pink trichrome blue/green. Elastic fiber disease Marfan fibrillin vs collagen disease EDS OI.

Decisive: silver black reticular III, Orcein black elastic, wavy pink bundles type I dense.

Compare & Distinguish

Clinical Meaning

Feature	Dense regular	Dense irregular	Loose
Fiber arrangement	Parallel type I	Interwoven type I	Random all components
Cells	Fibroblasts rows	Fibroblasts	Fibroblasts macrophages mast plasma
Location	Tendon ligament	Reticular dermis capsules	Papillary dermis around vessels
Function	Tensile one direction	Tensile many directions	Support diffusion immune
Feature	Collagen I	Collagen III reticular	Elastic fiber
---	---	---	---
Type	Fibrillar thick	Fibrillar thin type III	Elastin core + fibrillin
Stain	Trichrome blue/green thick	Silver black delicate	Orcein/Verhoeff brown-black

Feature	Dense regular	Dense irregular	Loose
Location	Skin tendon bone	Vessels lymphoid stroma	Large artery ligamentum nuchae
Disease	OI brittle	Vascular EDS	Marfan

Marfan: fibrillin-1 scaffold defect elastic weak aortic media cystic necrosis aneurysm lens dislocation arachnodactyly. Distinguish EDS collagen hypermobile.

Scurvy: vitamin C deficiency proline/lysine hydroxylation fails weak triple helix bleeding poor wound.

OI: type I collagen mutation brittle bones.

What to Remember

High-Yield

Must Know: Collagen I bone/skin/tendon OI, III reticular granulation vascular EDS, IV network BM Goodpasture/Alport. Elastic elastin core + fibrillin Marfan Orcein. GAGs water compression. Cells fibroblast builder macrophage defense mast metachromatic heparin/histamine plasma clock-face basophilic antibody.

Must Distinguish: Dense regular vs irregular vs loose, collagen vs elastic disease, reticular silver vs collagen trichrome vs elastic Orcein.

Must Understand: Hydroxylation vit C mechanism, why III→I wound healing, why ground substance GAGs hold water.

Common Misconceptions

Misconception: Collagen one type. Reality many types fibrillar I II III vs sheet-forming IV type determines location disease.

Misconception: Ground substance inert filler. Reality GAGs highly hydrated functional compression resistance and signaling.

Integrated Summary

Check Your Understanding

1. Explain scurvy mechanism from collagen biosynthesis.
2. Reconstruct collagen I III IV table with location disease.
3. Why Marfan aortic aneurysm while EDS hypermobile joints?
4. Distinguish dense regular vs irregular vs loose decisive features.
5. Sequence wound healing explain III→I switch.

Bridge

The framework is built. Next, a specialized framework that stores energy and signals: adipose tissue.

Rapid Review

- Components cells+fibers collagen I III IV elastic elastin+fibrillin reticular III + ground substance GAGs water.
- Collagen I bone/skin/tendon OI, III reticular granulation vascular EDS, IV network BM Goodpasture/Alport.
- Elastic elastin core+fibrillin Marfan Orcein.
- Cells fibroblast macrophage mast metachromatic plasma clock-face.
- CT types loose dense regular parallel dense irregular interwoven elastic reticular silver mucoid.

Chapter 6 — Adipose Tissue

Junqueira 17th Edition — Chapter 6

> White adipocytes store energy as a single large droplet and signal via leptin, while brown adipocytes generate heat via many small droplets, abundant mitochondria, and uncoupling protein 1.

Opening Question

How does fat do more than store energy — and why does its location decide metabolic risk?

Why This Matters

White adipose endocrine leptin adiponectin links obesity diabetes. Brown thermogenesis UCP1 therapeutic target. Lipoma vs liposarcoma distinction uses morphology.

Learning Objectives

By the end of this chapter, you will be able to:

1. Distinguish white unilocular vs brown multilocular vs beige by morphology mitochondria function.
2. Explain white endocrine and brown thermogenesis UCP1 mechanism.
3. Describe adipose as specialized connective tissue rich vascularity.
4. Predict metabolic consequence visceral vs subcutaneous white.

The Landscape

Adipose = connective tissue where lipid storage dominates but endocrine and thermogenic functions make metabolic organ.

The Core Principle

Lipid droplet size and mitochondria number determine function: storage vs heat.

> CORE IDEA: White adipocytes store energy as a single large droplet and signal via leptin, while brown adipocytes generate heat via many small droplets, abundant mitochondria, and uncoupling protein 1.

Building the System

White unilocular: single large droplet peripheral flattened nucleus signet-ring H&E lipid dissolved empty vacuole need frozen Oil Red O confirm ~up to ~100 µm scant mitochondria energy storage insulation cushioning leptin satiety adiponectin.

Brown multilocular: multiple small droplets central round nucleus many mitochondria dense cristae abundant capillaries UCP1 uncouples oxidative phosphorylation → heat not ATP thermogenesis neonate interscapular perirenal inducible cold sympathetic.

Beige/brite: inducible brown-like within white depots multilocular UCP1 positive cold/exercise induced.

Tissue organization: lobules separated septa dense irregular CT rich vascularity innervation sympathetic lipolysis/thermogenesis.

Why white signet-ring? One large droplet maximizes storage volume minimal surface. Why brown multilocular? Many small droplets increase surface for lipase and mitochondria proximity rapid oxidation heat.

Seeing and Distinguishing

Recognition Logic

Look for... large empty vacuole + peripheral nucleus white H&E empty because lipid dissolved. Then confirm with... Oil Red O red frozen. Do not confuse with... signet-ring carcinoma mucin not lipid atypia vs adipocyte benign. Decisive: multiple small droplets + central nucleus + eosinophilic granular cytoplasm mitochondria brown.

Compare & Distinguish

Clinical Meaning

Feature	White unilocular	Brown multilocular	Beige
Droplets	Single large	Many small	Many small induced
Nucleus	Peripheral flat	Central round	Central
Mitochondria	Few	Many UCP1	Many UCP1
Function	Storage endocrine leptin	Thermogenesis	Thermogenesis inducible
Location	Subcutaneous visceral marrow	Neonate interscapular adult perirenal	Within white

Obesity: visceral white more inflammatory insulin resistance subcutaneous less risky leptin deficiency hyperphagia.

Brown fat activation: cold sympathetic UCP1 heat potential obesity therapy.

Lipoma: benign white encapsulated. Liposarcoma atypical lipoblasts.

What to Remember

High-Yield

Must Know: White signet-ring peripheral nucleus single droplet leptin storage, brown multilocular central nucleus many mitochondria UCP1 heat, beige inducible. Lipid needs frozen Oil Red O H&E empty.

Must Distinguish: White vs brown vs beige adipocyte vs signet-ring carcinoma mucin vs lipid benign vs malignant atypia.

Must Understand: Why multilocular increases surface rapid lipolysis heat why UCP1 uncouples.

Common Misconceptions

Misconception: Fat inert. Reality endocrine leptin adiponectin cytokines.

Misconception: Brown only babies. Reality adult retains perirenal supraclavicular beige inducible.

Integrated Summary

Adipose specialized CT white unilocular storage + endocrine brown multilocular thermogenic UCP1 mitochondria beige inducible. Organization lobular vascularized. Recognition signet-ring empty H&E due lipid dissolution confirm frozen Oil Red O. Endocrine links obesity metabolic disease.

Check Your Understanding

1. Explain why H&E empty vacuole white adipocyte and how confirm lipid.
2. Compare white vs brown droplets nucleus mitochondria function UCP1.
3. Why brown many small droplets rather than one large?
4. Explain leptin adiponectin roles.

Bridge

Energy storage is organized. Next, tissue that lasts a lifetime and wires the body: nerve tissue and the nervous system.

Rapid Review

- White single droplet peripheral nucleus signet-ring storage leptin ~100 µm.
- Brown many droplets central nucleus many mitochondria UCP1 heat.
- Beige inducible brown-like in white.
- Frozen Oil Red O confirms lipid H&E empty.
- Lobules septa vascular.

Chapter 7 — Nerve Tissue & the Nervous System

Junqueira 17th Edition — Chapter 9

> In nerve tissue, dendrites receive signals, the axon conducts them, myelin insulates to speed conduction, nodes of Ranvier regenerate the action potential, and glia provide support, myelination, and barrier functions.

Opening Question

How does a cell that lasts a lifetime wire a body and still support itself?

Why This Matters

Multiple sclerosis CNS myelin, Guillain-Barré PNS myelin, Wallerian degeneration, blood-brain barrier explains why many drugs don't enter brain. Neuron post-mitotic glia support.

Learning Objectives

By the end of this chapter, you will be able to:

1. Describe neuron parts perikaryon Nissl dendrites axon hillock axon terminals and cytoskeleton neurofilament microtubule transport.
2. Compare PNS myelin one Schwann per internode neurilemma vs CNS myelin oligodendrocyte many internodes no neurilemma no basal lamina vs unmyelinated Schwann envelops many axons.
3. Identify glia astrocyte BBB foot processes oligodendrocyte myelin microglia phagocytosis ependyma and functions.
4. Explain synapse types and blood-brain barrier structure.

The Landscape

Neuron = long-lived polarized excitable cell dendrite→soma→axon→synapse; glia = support myelination barrier immune.

The Core Principle

Myelin speeds conduction saltatory jumping between nodes Ranvier; PNS one Schwann per internode with neurilemma basal lamina allows regeneration, CNS oligodendrocyte many internodes no neurilemma limits regeneration.

> CORE IDEA: In nerve tissue, dendrites receive signals, the axon conducts them, myelin insulates to speed conduction, nodes of Ranvier regenerate the action potential, and glia provide support, myelination, and barrier functions.

Building the System

Neuron: perikaryon Nissl RER basophilic prominent nucleolus dendrites tapering spines receive axon hillock origin no Nissl action potential initiation axon uniform diameter no RER neurofilaments + microtubules transport kinesin anterograde dynein retrograde terminals synaptic vesicles.

Cytoskeleton: neurofilaments intermediate stability diameter microtubules transport actin growth cone.

Synapse: chemical presynaptic active zone vesicles cleft postsynaptic density vs electrical gap junction chemical needs Ca^{2+}.

Myelin:

- PNS Schwann wraps plasma membrane concentrically one axon segment one internode per Schwann mesaxon compact myelin neurilemma Schwann cytoplasm + basal lamina outside node Ranvier bare axon Na^+ channels Schmidt-Lanterman incisures cytoplasm channels.
- CNS oligodendrocyte process wraps many axons many internodes per cell no neurilemma no basal lamina no incisures nodes.
- Unmyelinated PNS one Schwann envelops many small axons without wrapping no myelin still basal lamina.
- Function increases membrane resistance decreases capacitance saltatory conduction.

Glia: Astrocyte star GFAP foot processes around capillaries neurons BBB induction glutamate uptake scar. Oligodendrocyte small few processes myelin CNS. Microglia mesenchymal phagocytosis immune. Ependyma cuboidal ciliated lines ventricles CSF. Schwann satellite PNS glia.

Blood-brain barrier: continuous capillary endothelium tight junctions claudin + basal lamina + astrocyte foot processes no fenestrations pericytes.

Why PNS regenerates better than CNS? PNS Schwann basal lamina tube neurilemma guides regrowing axon after Wallerian degeneration secretes neurotrophins. CNS oligodendrocyte no basal lamina plus astrocyte scar myelin inhibitors Nogo block regrowth.

Why nodes Ranvier? Myelin insulates action potential jumps node to node Na^+ channels concentrated faster.

Seeing and Distinguishing

Recognition Logic

Look for... large cell Nissl basophilic perikaryon prominent nucleolus processes neuron. Small dark nuclei around glia.

Then confirm... myelin layers around axon Schwann cytoplasm outside basal lamina PNS myelin. Many axons myelinated one cell body no basal lamina CNS. Axons groups enveloped without wrapping unmyelinated.

Do not confuse with... Nissl vs plasma cell basophilia both RER but neuron processes large nucleus. Oligodendrocyte vs small lymphocyte both small dark but oligodendrocyte near myelin.

Decisive: neurilemma basal lamina one internode per Schwann PNS many internodes per oligodendrocyte no basal lamina CNS.

Compare & Distinguish

Clinical Meaning

Feature	PNS myelin	CNS myelin	Unmyelinated PNS
Cell	Schwann	Oligodendrocyte	Schwann
Internodes per cell	1	Many	None envelops many
Neurilemma + basal lamina	Yes	No	Yes basal lamina
Regeneration	Yes tube guides	Poor scar inhibitors	Yes
Incisures	Yes Schmidt-Lanterman	No	No
Glia	Origin	Function	Marker
---	---	---	---
Astrocyte	Neuroectoderm	BBB support scar	GFAP
Oligodendrocyte	Neuroectoderm	Myelin CNS many internodes	—
Microglia	Mesoderm monocyte	Phagocytosis immune	—
Ependyma	Neuroectoderm	Lines ventricles ciliated	—
Schwann	Neural crest	Myelin PNS one internode	—

MS: autoimmune CNS myelin loss oligodendrocyte slowed conduction no neurilemma poor regeneration plaques.

Guillain-Barré: autoimmune PNS myelin Schwann ascending paralysis can regenerate basal lamina tubes remain.

Wallerian degeneration: axon cut distal myelin axon degenerate Schwann proliferates basal lamina tube remains guides regrowth ~1 mm/day.

What to Remember

High-Yield

Must Know: Neuron perikaryon Nissl RER dendrites axon hillock no Nissl axon microtubule neurofilament transport kinesin anterograde dynein retrograde myelin PNS one Schwann per internode neurilemma basal lamina node Ranvier Schmidt-Lanterman CNS oligodendrocyte many internodes no neurilemma unmyelinated Schwann envelops many astrocyte GFAP BBB foot oligodendrocyte myelin microglia phagocytosis ependyma ciliated ventricles BBB continuous endothelium tight junctions basal lamina astrocyte foot.

Must Distinguish: PNS vs CNS myelin astrocyte vs oligodendrocyte vs microglia Nissl vs plasma cell basophilia.

Must Understand: Saltatory conduction why PNS regenerates better than CNS BBB structure.

Common Misconceptions

Misconception: Schwann and oligodendrocyte same. Reality one Schwann one internode neurilemma basal lamina oligodendrocyte many internodes no neurilemma no basal lamina.

Misconception: Neuron divides. Reality post-mitotic no centrioles no regeneration perikaryon only axon regrowth PNS.

Integrated Summary

Neuron polarized receiving dendrites synthetic perikaryon Nissl conducting axon neurofilament/microtubule transport synaptic terminals. Myelin speeds conduction saltatory PNS Schwann one internode neurilemma basal lamina Schmidt-Lanterman CNS oligodendrocyte many internodes no neurilemma no basal lamina unmyelinated Schwann envelops many. Glia astrocyte GFAP BBB oligodendrocyte myelin microglia phagocytosis ependyma ciliated. BBB continuous endothelium tight + basal lamina + astrocyte foot.

Check Your Understanding

1. Compare PNS vs CNS myelin by cell internodes per cell neurilemma basal lamina regeneration.
2. Explain saltatory conduction why nodes Na+ channels.
3. Why PNS axon regenerates while CNS does not?
4. Describe BBB layers why many antibiotics don't cross.

Bridge

Nerve needs supply and connection. Next, the system that supplies and connects all tissues: the circulatory system.

Rapid Review

- Neuron perikaryon Nissl RER prominent nucleolus dendrites hillock no Nissl axon neurofilament/microtubule kinesin anterograde dynein retrograde.
- Myelin PNS Schwann 1 internode neurilemma basal lamina Schmidt-Lanterman node CNS oligodendrocyte many internodes no neurilemma unmyelinated Schwann envelops many.
- Glia astrocyte GFAP BBB foot oligodendrocyte myelin CNS microglia phagocytosis ependyma ciliated.

• BBB continuous endothelium tight claudin basal lamina astrocyte foot pericyte.

Chapter 8 — The Circulatory System

Junqueira 17th Edition — Chapter 11

> In the circulatory system, elastic arteries recoil to maintain flow, muscular arteries regulate resistance, capillary type determines permeability, and veins provide capacitance with valves ensuring unidirectional flow.

Opening Question

How does one tube plan build an aorta that stretches and a capillary that leaks on purpose?

Why This Matters

Atherosclerosis intima disease, aneurysm media elastic defect, varicose veins valve failure, lymphedema lymphatic failure. Capillary type predicts protein leak.

Learning Objectives

By the end of this chapter, you will be able to:

1. Describe common 3-layer plan intima endothelium+subendothelial media smooth muscle+elastic adventitia CT and how ratio changes along vascular tree.
2. Distinguish elastic artery muscular artery arteriole capillary types continuous fenestrated discontinuous/sinusoid venule vein lymphatic.
3. Explain why elastic artery recoils and muscular artery vasoconstricts.
4. Predict where edema protein leak or immune trafficking occurs based on capillary type.

The Landscape

Vessel = intima + media + adventitia. Intima is defined by endothelium plus subendothelial connective tissue. Media determines mechanical behavior elastic recoil vs vasoconstriction. Adventitia anchors.

The Core Principle

Media composition = function: elastic lamellae for recoil, smooth muscle for resistance, fenestrations for exchange, valves for unidirectional flow.

> CORE IDEA: In the circulatory system, elastic arteries recoil to maintain flow, muscular arteries regulate resistance, capillary type determines permeability, and veins provide capacitance with valves ensuring unidirectional flow.

Building the System

Heart: endocardium endothelium+subendothelial CT+Purkinje myocardium cardiac muscle epicardium mesothelium+CT+fat. Valves dense irregular CT+endothelium.

Elastic artery aorta pulmonary: large lumen intima thick subendothelial media many elastic lamellae 40-70 + smooth muscle adventitia vasa vasorum. Function Windkessel stretch systole recoil diastole continuous flow.

Muscular artery brachial femoral: intima thin media many smooth muscle up to 40 layers few elastic lamellae prominent internal elastic lamina external elastic lamina. Function vasoconstriction/dilation.

Arteriole: small <0.5 mm 1-2 smooth muscle layers no elastic lamina resistance blood pressure regulation.

Capillary: smallest endothelium+basal lamina+pericytes. Types:

- Continuous tight junctions no fenestrations continuous basal lamina BBB muscle lung least permeable.
- Fenestrated pores with diaphragms except kidney glomerulus no diaphragm continuous basal lamina endocrine intestine kidney peritubular choroid permeable small molecules.
- Discontinuous/sinusoid large gaps discontinuous basal lamina liver spleen marrow permeable proteins cells.

Venule: postcapillary venule leaky immune trafficking pericytes.

Vein: large lumen thin media few smooth muscle thick adventitia valves endothelium+CT prevent backflow capacitance.

Lymphatic: larger than blood capillary no basal lamina or discontinuous overlapping endothelium primary valves anchoring filaments no pericytes valves drains to nodes.

Why aorta many elastic lamellae while muscular many smooth muscle? Aorta must recoil maintain diastolic flow elastic stores energy. Muscular must regulate flow organs smooth muscle constricts/dilates.

Why glomerular fenestrated without diaphragm while intestinal with diaphragm? Glomerulus needs high filtration fluid no diaphragm increases permeability. Intestine selective absorption diaphragm limits protein leak.

Seeing and Distinguishing

Recognition Logic

Look for... thick wall round lumen many wavy elastic lamellae elastic artery. Thick wall round lumen many smooth muscle prominent internal elastic muscular artery. Small wall 1-2 muscle arteriole. Single endothelium basal lamina pericyte capillary. Large irregular lumen thin wall valves vein. No basal lamina overlapping endothelium anchoring filaments lymphatic.

Then confirm with... internal elastic lamina prominent muscular many elastic lamellae elastic fenestrations EM capillary type.

Do not confuse with... artery vs vein artery round thick media vein large thin media valves. Blood capillary continuous basal lamina vs lymphatic no basal lamina.

Decisive: elastic lamellae many elastic artery muscle many + internal elastic muscular 1-2 muscle arteriole single endothelial capillary valves thin media vein.

Compare & Distinguish

Clinical Meaning

Feature	Elastic artery	Muscular artery	Arteriole	Vein
Lumen	Large	Medium	Small	Large irregular
Intima	Thick subendothelial	Thin internal elastic prominent	Thin	Thin
Media	Many elastic lamellae+muscle	Many muscle few elastic	1-2 muscle	Few muscle thin
Adventitia	Vasa vasorum	Vasa vasorum	None	Thick
Function	Recoil Windkessel	Vasoconstriction	Resistance BP	Capacitance valves
Capillary type	Fenestration	Basal lamina	Location	Permeability
Continuous	No	Continuous	BBB muscle lung skin	Low
Fenestrated with diaphragm	Yes with diaphragm	Continuous	Endocrine intestine peritubular	Moderate
Fenestrated without diaphragm	Yes no diaphragm	Continuous	Glomerulus	High fluid
Discontinuous sinusoid	Large gaps	Discontinuous	Liver spleen marrow	High protein/cells

Atherosclerosis: intima disease subendothelial LDL accumulation smooth muscle migration foam cells.

Aneurysm: media elastic defect Marfan fibrillin weak recoil dilation.

Varicose veins: valve failure backflow dilation.

What to Remember

High-Yield

Must Know: 3 layers intima endothelium+subendothelial media muscle+elastic adventitia CT. Elastic many elastic lamellae Windkessel muscular many muscle internal elastic vasoconstriction arteriole 1-2 muscle resistance capillary types continuous tight BBB fenestrated with diaphragm endocrine/intestine fenestrated without diaphragm glomerulus discontinuous sinusoid liver/spleen/marrow protein/cells vein large thin media valves capacitance lymphatic no basal lamina overlapping anchoring filaments valves.

Must Distinguish: Elastic vs muscular vs arteriole vs vein vs lymphatic capillary types.

Must Understand: Why media composition determines function why capillary type predicts permeability trafficking.

Common Misconceptions

Misconception: All capillaries same. Reality three types different permeability determines where protein leaks and where blood cells exit.

Misconception: Veins have thick media. Reality thin media thick adventitia valves capacitance.

Integrated Summary

Circulatory 3-layer plan modified intima endothelium always media elastic recoil vs muscle vasoconstriction adventitia anchor. Elastic aorta recoil muscular vasoregulation arteriole resistance capillary exchange type continuous tight BBB vs fenestrated diaphragm endocrine vs no diaphragm glomerulus vs discontinuous sinusoid liver/spleen/marrow vein capacitance valves lymphatic no basal lamina drainage. Structure predicts permeability disease.

Check Your Understanding

1. Explain Windkessel function from elastic lamellae.
2. Compare capillary types by fenestration basal lamina location permeability.
3. Distinguish artery vs vein vs lymphatic decisive features.
4. Why glomerular capillary fenestrations without diaphragm?

Bridge

The circulatory system carries a specialized connective tissue: blood.

Rapid Review

- Layers intima endothelium+subendothelial media muscle+elastic adventitia CT vasa vasorum.
- Elastic many lamellae recoil muscular many muscle internal elastic vasoconstriction arteriole 1-2 muscle resistance.
- Capillary continuous tight BBB fenestrated diaphragm endocrine/intestine fenestrated no diaphragm glomerulus discontinuous sinusoid liver/spleen/marrow.
- Vein large thin media valves capacitance lymphatic no BM overlapping anchoring filaments.

Chapter 9 — Blood

Junqueira 17th Edition — Chapter 12

> In blood, multilobed neutrophils are short-lived innate defenders, round dark small lymphocytes are long-lived adaptive cells, kidney-shaped monocytes are macrophage precursors, and biconcave discs without nuclei are red cells specialized for gas exchange.

Opening Question

How can you tell a bacterial infection from a parasitic one from a single blood smear?

Why This Matters

Neutrophilia bacterial eosinophilia parasite/allergy lymphocytosis viral monocytosis chronic infection left shift band neutrophils. RBC morphology anemia.

Learning Objectives

By the end of this chapter, you will be able to:

1. Describe RBC biconcave disc ~7.5 µm no nucleus no organelles eosinophilic why shape matters.
2. Distinguish neutrophil ~12-15 µm 3-5 lobes neutrophilic granules eosinophil bilobed large eosinophilic granules basophil bilobed obscured basophilic metachromatic small lymphocyte ~6-8 µm round dark nucleus scant cytoplasm monocyte ~12-20 µm kidney-shaped nucleus by morphology function.
3. Explain why nucleus shape + granule color predicts function.
4. Recognize platelet discoid no nucleus granules.

The Landscape

Blood = plasma + formed elements. Leukocyte morphology predicts defense neutrophil bacterial eosinophil parasite/allergy basophil heparin/histamine lymphocyte adaptive monocyte macrophage precursor.

The Core Principle

Leukocyte morphology reflects function, but requires integration of nucleus shape, granule content, size, and clinical context.

> CORE IDEA: In blood, multilobed neutrophils are short-lived innate defenders, round dark small lymphocytes are long-lived adaptive cells, kidney-shaped monocytes are macrophage precursors, and biconcave discs without nuclei are red cells specialized for gas exchange.

Building the System

RBC: biconcave disc ~7.5 µm diameter ~2 µm thick no nucleus no organelles eosinophilic hemoglobin flexible increases surface gas exchange lifespan ~120 days removed spleen.

Neutrophil: ~12-15 µm 3-5 lobes more lobes older neutrophilic fine granules specific+azurophilic bacterial phagocytosis pus first responder lifespan hours-days band horseshoe immature left shift.

Eosinophil: ~12-15 µm bilobed spectacles large eosinophilic specific granules major basic protein parasite defense allergy lifespan days Charcot-Leyden.

Basophil: ~12-15 µm bilobed granules obscure nucleus basophilic metachromatic granules heparin+histamine allergy anaphylaxis least numerous.

Small lymphocyte: ~6-8 µm round dark heterochromatic nucleus scant basophilic cytoplasm T vs B not distinguishable morphology need IHC recirculates long-lived memory.

Large lymphocyte/activated: ~10-15 µm more cytoplasm euchromatic immunoblast.

Monocyte: ~12-20 µm largest leukocyte kidney/horseshoe nucleus ground-glass cytoplasm lysosomes precursor macrophage chronic infection antigen presentation.

Platelet: discoid ~2-3 µm no nucleus granules alpha+dense clotting derived megakaryocyte.

Why RBC biconcave disc no nucleus? Biconcave increases surface area gas exchange flexibility squeeze through capillaries. No nucleus/organelles more space hemoglobin.

Why neutrophil multilobed? Lobes allow squeezing through endothelium diapedesis.

Seeing and Distinguishing

Recognition Logic

Look for... nucleus shape first multilobed 3-5 neutrophil bilobed spectacles red granules eosinophil bilobed obscured dark purple granules basophil round dark scant cytoplasm small lymphocyte kidney monocyte no nucleus biconcave RBC.

Then confirm with... granule color eosinophilic red eosinophil basophilic purple basophil neutrophilic fine neutrophil.

Do not confuse with... band horseshoe vs segmented lobed neutrophil band immature left shift infection. Small lymphocyte vs RBC lymphocyte has nucleus dark RBC no nucleus.

Decisive: nucleus shape + granule color.

Compare & Distinguish

Clinical Meaning

Cell	Size	Nucleus	Granules	Function
RBC	~7.5 µm	None	None eosinophilic	O2/CO2
Neutrophil	~12-15 µm	3-5 lobes	Neutrophilic fine	Bacterial phagocytosis
Eosinophil	~12-15 µm	Bilobed spectacles	Large eosinophilic	Parasite allergy
Basophil	~12-15 µm	Bilobed obscured	Basophilic metachromatic	Heparin histamine
Small lymphocyte	~6-8 µm	Round dark	Scant basophilic	Adaptive T/B
Monocyte	~12-20 µm	Kidney	Ground-glass	Macrophage precursor
Platelet	~2-3 µm	None	Granules	Clotting

Left shift: bands increased bacterial infection.

Eosinophilia: parasite allergy asthma.

Lymphocytosis: viral chronic lymphocytic leukemia.

What to Remember

High-Yield

Must Know: RBC ~7.5 µm biconcave no nucleus eosinophilic neutrophil ~12-15 µm 3-5 lobes bacterial eosinophil bilobed large eosinophilic parasite basophil bilobed obscured metachromatic heparin histamine small lymphocyte ~6-8 µm round dark scant monocyte ~12-20 µm kidney macrophage precursor platelet ~2-3 µm no nucleus clotting.

Must Distinguish: Neutrophil vs eosinophil vs basophil lobes + granule color small lymphocyte vs RBC band vs segmented.

Must Understand: Why shape predicts function why nucleus shape matters diapedesis.

Common Misconceptions

Misconception: Lymphocyte T vs B distinguishable morphology. Reality not distinguishable routine H&E need IHC CD markers.

Misconception: Platelet is cell. Reality fragment megakaryocyte no nucleus.

Integrated Summary

Blood circulating CT RBC biconcave disc no nucleus gas exchange neutrophil 3-5 lobes bacterial eosinophil bilobed eosinophilic parasite basophil bilobed obscured basophilic allergy small lymphocyte round dark adaptive monocyte kidney macrophage precursor platelet fragment clotting. Nucleus + granule = function.

Check Your Understanding

1. How to tell bacterial vs parasitic infection from smear?
2. Distinguish neutrophil vs eosinophil vs basophil decisive features.
3. Why RBC biconcave no nucleus?
4. What is left shift and what does it indicate?

Bridge

Blood cells die in days to months. Where do they come from? Next, the marrow factory: hemopoiesis.

Rapid Review

- RBC ~7.5 μm biconcave no nucleus eosinophilic 120d.
- Neutrophil ~12-15 μm 3-5 lobes neutrophilic bacterial band left shift.
- Eosinophil bilobed large eosinophilic parasite allergy.
- Basophil bilobed obscured metachromatic heparin histamine.
- Small lymphocyte ~6-8 μm round dark scant T/B not morphologically distinct.
- Monocyte ~12-20 μm kidney macrophage precursor chronic.
- Platelet ~2-3 μm fragment clotting.

Chapter 10 — Hemopoiesis

Junqueira 17th Edition — Chapter 13

> In hemopoiesis, early erythroid precursors are basophilic due to abundant rough endoplasmic reticulum, intermediate forms are polychromatic as hemoglobin accumulates, and late forms are eosinophilic as hemoglobin dominates.

Opening Question

How does marrow make 200 billion red cells a day without making a tumor?

Why This Matters

Aplastic anemia niche failure leukemia maturation arrest EPO G-CSF therapy reticulocyte count indicates marrow response.

Learning Objectives

By the end of this chapter, you will be able to:

1. Describe marrow organization red vs yellow sinusoids stromal niche why sinusoids needed.
2. Sequence erythropoiesis proerythroblast basophilic → polychromatic → orthochromatic → reticulocyte → RBC and explain basophilia→eosinophilia shift.
3. Sequence granulopoiesis myeloblast → promyelocyte azurophilic → myelocyte specific → metamyelocyte → band → segmented and megakaryopoiesis endomitosis platelet demarcation.
4. Explain growth factors EPO G-CSF GM-CSF thrombopoietin.

The Landscape

Hemopoiesis hierarchical differentiation specialized niche stem → progenitors → precursors morphology reflecting synthesis activity released via sinusoids.

The Core Principle

Niche + growth factor + morphology reflecting synthesis = regulated production.

> CORE IDEA: In hemopoiesis, early erythroid precursors are basophilic due to abundant rough endoplasmic reticulum, intermediate forms are polychromatic as hemoglobin accumulates, and late forms are eosinophilic as hemoglobin dominates.

Building the System

Marrow: red active hematopoietic cords + sinusoids fenestrated discontinuous vs yellow inactive adipocytes. Stromal cells macrophages nurse sinusoids allow mature cells enter blood but retain immature.

Stem cell: pluripotent self-renewal rare.

Progenitors: myeloid vs lymphoid.

Erythropoiesis: proerythroblast large basophilic RER → basophilic erythroblast → polychromatic basophilic+eosinophilic hemoglobin mixed → orthochromatic eosinophilic pyknotic nucleus → reticulocyte no nucleus residual RER reticulum supravital stain → RBC biconcave no nucleus. Nucleus expelled organelles lost.

Granulopoiesis: myeloblast large euchromatic no granules → promyelocyte azurophilic primary granules lysosomes → myelocyte specific secondary granules neutrophilic/eosinophilic/basophilic → metamyelocyte kidney nucleus → band horseshoe → segmented lobed. Myelocyte last dividing stage.

Megakaryopoiesis: megakaryoblast → megakaryocyte large polyploid endomitosis DNA replication without division → platelet demarcation membranes → platelets released via sinusoids.

Growth factors: EPO kidney → RBC G-CSF → neutrophil GM-CSF → granulocyte/macrophage thrombopoietin liver → platelet IL-3 SCF.

Why reticulocyte still has RER? Recently enucleated still needs synthesize hemoglobin residual RER visible supravital stain brilliant cresyl blue indicates marrow response increased hemolysis.

Why sinusoids discontinuous? Allows mature cells squeeze through but barrier immature.

Seeing and Distinguishing

Recognition Logic

Look for... basophilic cytoplasm large nucleus early erythroid eosinophilic pyknotic late kidney nucleus granules metamyelocyte large polyploid demarcation megakaryocyte.

Decisive: basophilic→eosinophilic shift erythroid azurophilic→specific granuloid polyploid megakaryocyte.

Compare & Distinguish

Clinical Meaning

Lineage	Early	Mid	Late	Decisive
Erythroid	Basophilic RER	Polychromatic mixed	Orthochromatic eosinophilic pyknotic → reticulocyte	Basophilic→eosinophilic shift
Granuloid	Myeloblast no granules	Promyelocyte azurophilic primary	Myelocyte specific last division	Granule type
Megakaryoid	Small	Large polyploid	Platelet demarcation	Endomitosis

Aplastic anemia: niche failure pancytopenia fatty marrow.

Leukemia: maturation arrest blasts increased.

Reticulocytosis: hemolysis bleeding marrow compensates increased reticulocytes.

What to Remember

High-Yield

Must Know: Red vs yellow marrow sinusoids discontinuous erythropoiesis basophilic→polychromatic→orthochromatic→reticulocyte→RBC granulopoiesis myeloblast→promyelocyte azurophilic→myelocyte specific last division→metamyelocyte→band→segmented megakaryocyte endomitosis polyploid demarcation platelets EPO RBC G-CSF neutrophil TPO platelet.

Must Distinguish: Basophilic vs polychromatic vs orthochromatic erythroblast promyelocyte vs myelocyte reticulocyte vs RBC.

Must Understand: Why morphology reflects synthesis why sinusoids needed why reticulocyte count indicates marrow response.

Common Misconceptions

Misconception: Reticulocyte immature with nucleus. Reality no nucleus residual RER reticulum supravital.

Misconception: All marrow red. Reality adult yellow fatty long bones red flat bones sternum iliac crest biopsy site.

Integrated Summary

Marrow niche stromal + sinusoids discontinuous allows regulated release. Erythropoiesis basophilic RER → polychromatic mixed → eosinophilic hemoglobin → enucleation → reticulocyte RER residual → RBC. Granulopoiesis myeloblast no granules → promyelocyte azurophilic → myelocyte specific last division → metamyelocyte kidney → band horseshoe → segmented lobed. Megakaryopoiesis endomitosis polyploid demarcation platelets. Growth factors EPO G-CSF TPO regulate.

Check Your Understanding

1. Sequence erythropoiesis explain color shift.
2. Why myelocyte last dividing stage important?
3. Explain reticulocytosis meaning.
4. Compare red vs yellow marrow why iliac crest biopsy site.

Bridge

New lymphocytes need education before defense. Next, primary versus secondary lymphoid organs and T versus B geography.

Rapid Review

- Marrow red active cords+sinusoids discontinuous vs yellow fatty.
- Erythropoiesis basophilic→polychromatic→orthochromatic eosinophilic pyknotic→reticulocyte RER residual→RBC ~7.5 μm.
- Granulopoiesis myeloblast→promyelocyte azurophilic primary→myelocyte specific secondary last division→metamyelocyte kidney→band→segmented.
- Megakaryocyte endomitosis polyploid demarcation platelets.
- Growth factors EPO RBC G-CSF neutrophil TPO platelet reticulocyte count marrow response.

Chapter 11 — The Immune System & Lymphoid Organs

Junqueira 17th Edition — Chapter 14

> Primary lymphoid organs, marrow and thymus, produce naive lymphocytes, while secondary organs, nodes, spleen, and MALT, are where lymphocytes encounter antigen. In secondary organs, the periarteriolar lymphoid sheath is T-cell rich and follicles are B-cell rich.

Opening Question

Why does losing one gland in a newborn's chest leave them defenseless against viruses for life?

Why This Matters

DiGeorge no thymus → T deficiency viral/fungal + hypocalcemia parathyroid follicular hyperplasia B vs paracortical T hyperplasia splenectomy Howell-Jolly bodies encapsulated bacteria risk SCID.

Learning Objectives

By the end of this chapter, you will be able to:

1. Classify primary bone marrow B thymus T vs secondary lymph node spleen MALT and functions education vs execution.
2. Describe thymus cortex vs medulla Hassall corpuscles blood-thymus barrier why no germinal centers.
3. Describe lymph node cortex B follicles paracortex T HEV medulla cords/sinuses lymph flow direction.
4. Describe spleen white pulp PALS T + follicles B + central artery red pulp cords + sinusoids why filters blood no afferent lymphatics.
5. Explain MALT Peyer patches tonsils M cells antigen sampling.

The Landscape

Immune organized education vs execution and T vs B geography. Every secondary organ has T zone and B zone recognizing zones is recognizing organ.

The Core Principle

Primary organs make naive lymphocytes secondary organs where antigen encountered. T vs B geography diagnostic.

> CORE IDEA: Primary lymphoid organs, marrow and thymus, produce naive lymphocytes, while secondary organs, nodes, spleen, and MALT, are where lymphocytes encounter antigen. In secondary organs, the periarteriolar lymphoid sheath is T-cell rich and follicles are B-cell rich.

Building the System

Primary: Bone marrow B maturation negative selection. Thymus T maturation cortex immature T densely packed dark blood-thymus barrier continuous endothelium basal lamina pericyte epithelial reticular cell prevents antigen entry medulla lighter Hassall corpuscles keratinized epithelial reticular whorls + mature T negative selection no germinal centers no B response involutes with age myoid cells.

Secondary: Lymph node capsule trabeculae cortex B follicles primary no germinal center vs secondary germinal center light+dark zones B proliferation paracortex T + high endothelial venules HEV cuboidal endothelium lymphocyte homing medulla cords plasma cells + sinuses macrophages. Lymph flow afferent → subcapsular sinus → cortical sinuses → medullary sinuses → efferent filters lymph.

Spleen capsule trabeculae white pulp PALS T around central artery + follicles B attached marginal zone B red pulp cords macrophages plasma + sinusoids discontinuous endothelium no afferent lymphatics filters blood open vs closed circulation Howell-Jolly bodies after splenectomy nuclear remnants RBC not removed.

MALT mucosa-associated Peyer patches ileum B follicles + T interfollicular + M cells microfold sample antigen no microvilli intraepithelial lymphocytes tonsils crypts palatine stratified squamous non-keratinized + crypts pharyngeal lingual.

Why thymus blood-thymus barrier and no germinal centers? Barrier prevents premature antigen exposure during T education germinal centers B response thymus educates T so no follicles. Hassall corpuscles keratinized epithelial remnants.

Why spleen no afferent lymphatics? Filters blood not lymph so antigen arrives via blood.

Why HEV cuboidal? Allows lymphocyte homing from blood to paracortex T zone.

Seeing and Distinguishing

Recognition Logic

Look for... Hassall corpuscles + no germinal centers + cortex dark medulla light thymus. Follicles cortex + paracortex + medulla cords/sinuses + subcapsular sinus lymph node. PALS around central artery + follicles + red pulp cords/sinusoids + no subcapsular sinus spleen. Crypts + stratified squamous + lymphoid follicles tonsil.

Then confirm... HEV cuboidal node paracortex T central artery spleen PALS T.

Do not confuse with... thymus vs node thymus no germinal centers Hassall present. Spleen vs node spleen no afferent has red pulp central artery. PALS T vs follicle B.

Decisive: Hassall thymus germinal centers + HEV + subcapsular sinus node PALS + central artery + red pulp spleen M cells MALT.

Compare & Distinguish

Clinical Meaning

Organ	T zone	B zone	Special	Filters
Thymus	Cortex immature + medulla mature	None no germinal centers	Hassall blood-thymus barrier	Blood education
Lymph node	Paracortex HEV	Cortex follicles germinal centers	Subcapsular sinus afferent/efferent	Lymph
Spleen	PALS central artery	Follicles attached PALS	No afferent red pulp cords/sinusoids	Blood
MALT Peyer	Interfollicular T	Follicles	M cells no capsule	Antigen lumen

DiGeorge 22q11.2 deletion: no thymus + parathyroids → T deficiency viral/fungal + hypocalcemia heart defects.

Follicular hyperplasia: B follicles enlarged bacterial infection.

Paracortical hyperplasia: T zone enlarged viral.

What to Remember

High-Yield

Must Know: Primary marrow B thymus T education secondary node spleen MALT execution. Thymus cortex dark immature blood-thymus barrier continuous endothelium+basal lamina+epithelial reticular medulla Hassall keratinized no germinal centers. Node cortex B follicles primary/secondary germinal center paracortex T HEV cuboidal medulla cords plasma sinuses macrophages lymph flow afferent→subcapsular→cortical→medullary→efferent filters lymph. Spleen white pulp PALS T central artery + follicles B red pulp cords+sinusoids discontinuous no afferent filters blood marginal zone. MALT Peyer B follicles T interfollicular M cells sample tonsil crypts.

Must Distinguish: Thymus vs node vs spleen vs MALT PALS T vs follicle B follicular vs paracortical hyperplasia.

Must Understand: Why thymus barrier and no germinal centers why spleen no afferent why HEV cuboidal homing.

Common Misconceptions

Misconception: Spleen is primary. Reality secondary filters blood no afferent.

Misconception: Thymus has germinal centers. Reality no because T education not B response.

Integrated Summary

Immune education vs execution marrow B + thymus T cortex immature barrier medulla Hassall negative selection no germinal centers make naive node cortex B follicles paracortex T HEV medulla cords/sinuses afferent→efferent filters lymph spleen white pulp PALS T central artery + follicles B red pulp cords/sinusoids filters blood no afferent MALT Peyer M cells tonsil crypts use them. T vs B geography diagnostic filters determine antigen source.

Check Your Understanding

1. Why thymus no germinal centers and blood-thymus barrier?
2. Trace lymph flow through node explain why subcapsular sinus first.
3. Compare spleen vs node filters afferent T/B zones special features.
4. Explain DiGeorge triad embryology.
5. Why splenectomy causes Howell-Jolly bodies and encapsulated bacteria risk?

Bridge

Defense protects the gut. The gut is the largest interface with the outside world. Next, how one tube digests, absorbs, and protects itself.

Rapid Review

- Primary marrow B thymus T cortex dark barrier medulla Hassall no germinal centers.
- Node cortex B follicles germinal centers paracortex T HEV medulla cords plasma sinuses macrophages flow afferent→subcapsular→cortical→medullary→efferent filters lymph.
- Spleen white PALS T central artery + follicles B red cords+sinusoids discontinuous no afferent filters blood Howell-Jolly after splenectomy.
- MALT Peyer B follicles T interfollicular M cells tonsil crypts stratified squamous.
- DiGeorge no thymus T deficiency + hypocalcemia follicular B hyperplasia bacterial paracortical T viral.

Chapter 12 — Digestive Tract

Junqueira 17th Edition — Chapter 15

> The digestive tract shares a common four-layer plan — mucosa, submucosa, muscularis externa, and serosa or adventitia — with regional modifications in epithelium, glands, villi, lymphoid tissue, and muscle that explain specialized functions.

Opening Question

How does one tube digest a steak, absorb fat, and protect itself from its own acid?

Why This Matters

Barrett metaplasia, peptic ulcer HCl, celiac villus atrophy, Crohn vs ulcerative colitis, Hirschsprung aganglionosis (no Auerbach/Meissner).

Learning Objectives

By the end of this chapter, you will be able to:

1. State common 4-layer plan mucosa epithelium+lamina propria+muscularis mucosae submucosa dense irregular+Meissner plexus muscularis externa inner circular+outer longitudinal+Auerbach plexus serosa/adventitia and how modified regionally.

2. Distinguish esophagus stomach cardia/fundus/body/pylorus cells parietal HCl/intrinsic factor chief pepsinogen mucous neck enteroendocrine small intestine villi/crypts duodenum Brunner jejunum ileum Peyer large intestine anal canal.

3. Explain why duodenum has Brunner glands and ileum has Peyer patches.

4. Predict consequence muscularis mucosae contraction and Auerbach plexus loss.

The Landscape

GI tract = common layered plan modified regionally for specialized function protection esophagus stratified squamous acid secretion stomach absorption small intestine villi water absorption colon.

The Core Principle

4 layers + regional specialization: epithelium type + glands + villi + lymphoid + muscle thickness = function.

> CORE IDEA: The digestive tract shares a common four-layer plan — mucosa, submucosa, muscularis externa, and serosa or adventitia — with regional modifications in epithelium, glands, villi, lymphoid tissue, and muscle that explain specialized functions.

Building the System

Common plan: Mucosa epithelium type varies + lamina propria loose CT + glands + immune + muscularis mucosae thin smooth muscle moves mucosa. Submucosa dense irregular CT + submucosal plexus Meissner parasympathetic ganglia secretion. Muscularis externa inner circular + outer longitudinal smooth muscle + myenteric plexus Auerbach between motility. Serosa peritoneal cavity mesothelium+CT vs adventitia no mesothelium dense CT.

Esophagus: stratified squamous non-keratinized protection submucosal glands mucous muscularis externa upper skeletal + middle mixed + lower smooth adventitia no serosa except abdominal.

Stomach: rugae folds pits foveolae + glands. Cardia mucous glands. Fundus/body pits shallow glands with parietal eosinophilic central HCl via H+/K+ ATPase + intrinsic factor B12 chief basophilic basal RER + eosinophilic apical pepsinogen mucous neck mucin enteroendocrine hormones. Pylorus deep pits mucous + G cells gastrin. Muscularis externa 3 layers oblique+circular+longitudinal.

Small intestine: plicae circulares mucosa+submucosa + villi mucosa finger-like epithelium simple columnar absorptive enterocytes microvilli brush border + goblet + crypts Lieberkühn stem + Paneth lysozyme/defensins + enteroendocrine. Lamina propria lacteal lymphatic fat absorption. Duodenum Brunner glands submucosa alkaline mucus protects from acid villi leaf-like many goblet. Jejunum tall villi many plicae few goblet no Brunner no Peyer. Ileum short villi Peyer patches MALT B follicles + M cells few plicae.

Large intestine: no villi no plicae crypts goblet many taenia coli 3 bands outer longitudinal haustra appendix lymphoid + mesoappendix.

Anal canal: upper columnar + lower stratified squamous anal columns internal sphincter smooth circular external skeletal.

Why duodenum Brunner glands submucosa? Duodenum receives acidic chyme stomach Brunner alkaline mucus protects epithelium neutralizes acid pancreatic enzymes.

Why ileum Peyer patches? Ileum near colon bacteria immune sampling M cells.

Why villi small intestine not colon? Small needs absorption surface colon water absorption no need villi.

Seeing and Distinguishing

Recognition Logic

Look for... stratified squamous + submucosal glands esophagus. Pits + parietal eosinophilic central + chief basophilic stomach fundus. Villi + crypts + Brunner submucosa duodenum. Tall villi no Brunner no Peyer jejunum. Villi short + Peyer patches ileum. No villi + crypts + many goblet + taenia colon.

Then confirm... parietal HCl intrinsic factor eosinophilic chief pepsinogen basophilic Paneth eosinophilic granules lysozyme crypt base goblet pale mucin Brunner mucous submucosa.

Do not confuse with... duodenum vs jejunum vs ileum duodenum Brunner jejunum tall villi no Peyer ileum Peyer. Stomach vs intestine stomach no villi pits parietal/chief.

Decisive: Brunner duodenum Peyer ileum parietal+chief fundus no villi many goblet taenia colon.

Compare & Distinguish

Clinical Meaning

Feature	Esophagus	Stomach fundus	Duodenum	Jejunum	Ileum	Colon
Epithelium	Stratified squamous non-kerat	Simple columnar mucous surface	Simple columnar + goblet	Simple columnar + goblet	Simple columnar + goblet	Simple columnar + goblet many
Villi	No	No	Yes leaf + Brunner submucosa	Yes tall	Yes short + Peyer	No
Glands	Submucosal mucous	Gastric parietal chief mucous neck	Brunner submucosa	Crypts Paneth	Crypts Paneth + Peyer	Crypts no Paneth
Muscularis externa	Skeletal→smooth	3 layers	2 layers	2 layers	2 layers	Taenia coli 3 bands

Barrett: esophagus stratified squamous → columnar goblet acid metaplasia premalignant.

Peptic ulcer: parietal HCl excess H. pylori.

Celiac: villus atrophy gluten loss absorption.

What to Remember

High-Yield

Must Know: 4 layers mucosa epithelium+lamina propria+muscularis mucosae submucosa Meissner muscularis externa inner circular outer longitudinal Auerbach serosa/adventitia. Esophagus stratified squamous submucosal glands skeletal→smooth. Stomach pits + glands cardia mucous fundus parietal HCl intrinsic factor eosinophilic chief pepsinogen basophilic mucous neck enteroendocrine pylorus G gastrin 3 muscle layers. Small intestine plicae+villi+crypts Paneth lysozyme enteroendocrine lacteal duodenum Brunner alkaline leaf villi jejunum tall villi ileum Peyer M cells. Colon no villi crypts many goblet taenia.

Must Distinguish: Duodenum vs jejunum vs ileum vs colon vs esophagus vs stomach parietal vs chief Brunner vs Peyer.

Must Understand: Why Brunner protects acid why Peyer near colon bacteria why villi only small intestine absorption why muscularis mucosae moves mucosa.

Common Misconceptions

Misconception: Stomach has villi. Reality no villi has pits rugae villi only small intestine.

Misconception: Brunner in jejunum. Reality duodenum only submucosa.

Misconception: Colon has Paneth. Reality no Paneth except appendix mainly small intestine crypts.

Integrated Summary

GI tract common 4-layer plan mucosa epithelium+lamina propria+muscularis mucosae + submucosa dense irregular+Meissner secretion + muscularis externa inner circular outer longitudinal+Auerbach motility + serosa/adventitia modified regionally esophagus stratified squamous protection submucosal mucous stomach pits glands parietal HCl intrinsic factor chief pepsinogen mucous neck G gastrin 3 muscle small intestine plicae villi crypts Paneth lacteal absorption duodenum Brunner alkaline ileum Peyer immune colon no villi many goblet taenia water.

Check Your Understanding

1. State 4 layers and plexuses functions.
2. Compare duodenum vs jejunum vs ileum decisive features.
3. Explain why duodenum needs Brunner and ileum needs Peyer.
4. Distinguish parietal vs chief stain function.
5. What happens if Auerbach plexus absent?

Bridge

The tube needs accessory organs to process nutrients. Next, organs associated with the digestive tract: liver, pancreas, and salivary glands.

Rapid Review

- 4 layers mucosa epithelium+lamina+muscularis mucosae submucosa Meissner muscularis externa circular+longitudinal Auerbach serosa/adventitia.
- Esophagus stratified squamous submucosal glands.
- Stomach pits glands fundus parietal HCl intrinsic eosinophilic + chief pepsinogen basophilic + mucous neck + enteroendocrine pylorus G gastrin.
- Small plicae villi crypts Paneth lacteal duodenum Brunner submucosa alkaline jejunum tall villi ileum Peyer M cells.
- Colon no villi crypts goblet many taenia Hirschsprung no ganglia megacolon.

Chapter 13 — Organs Associated with the Digestive Tract

Junqueira 17th Edition — Chapter 16

> The liver can be described by three models: the classic lobule centered on the central vein for blood flow, the portal lobule centered on the portal triad for bile drainage, and the acinus for metabolic zonation and injury patterns.

Opening Question

How does liver do 500 jobs with only a few cell types — and why does its blood flow make it vulnerable?

Why This Matters

Cirrhosis fibrosis disrupts sinusoids diabetes beta loss Sjögren salivary jaundice bilirubin hepatitis.

Learning Objectives

By the end of this chapter, you will be able to:

1. Describe liver organization classic lobule central vein portal lobule bile drainage acinus metabolic zones 1-3 why three models needed.
2. Identify hepatocyte plates sinusoids fenestrated discontinuous Kupffer macrophage Ito stellate vitamin A bile canaliculi portal triad.
3. Compare pancreas exocrine serous acinar vs endocrine islet alpha glucagon peripheral beta insulin central delta somatostatin.
4. Distinguish salivary serous vs mucous vs mixed acini and duct system intercalated→striated→excretory and myoepithelial.

The Landscape

Liver = epithelial cords + sinusoids three models explain different functions. Pancreas = exocrine protein secretion + endocrine glucose control. Salivary = serous protein vs mucous mucin.

The Core Principle

Liver cords + sinusoids + portal triad + 3 models; pancreas serous acinar basophilic basal eosinophilic apical + islet alpha peripheral beta central; salivary serous vs mucous + striated duct ion modification.

> CORE IDEA: The liver can be described by three models: the classic lobule centered on the central vein for blood flow, the portal lobule centered on the portal triad for bile drainage, and the acinus for metabolic zonation and injury patterns.

Building the System

Liver: Hepatocyte polygonal central nucleus eosinophilic mitochondria basophilic RER patches peroxisomes glycogen plates one cell thick radiating central vein. Sinusoids fenestrated discontinuous endothelium discontinuous basal lamina Kupffer macrophages phagocytosis Ito stellate cells vitamin A storage fibrosis collagen when activated. Space Disse between hepatocyte sinusoid microvilli plasma. Bile canaliculi between hepatocytes tight junctions seal drain canals Hering → bile ductules → interlobular bile ducts portal triad. Portal triad portal vein large thin hepatic artery small thick bile duct cuboidal epithelium. Models classic lobule hexagon central vein portal triads corners blood portal+arterial → central. Portal lobule triangle portal triad center bile drains center exocrine function. Acinus diamond portal triad + central vein corners zones 1 periportal oxygen rich vulnerable bile 2 mid 3 perivenular oxygen poor vulnerable ischemia centrilobular necrosis.

Gallbladder: mucosa simple columnar + lamina propria no muscularis mucosae muscle smooth serosa/adventitia concentrates bile.

Pancreas: Exocrine serous acinar basophilic basal RER + eosinophilic apical zymogen granules amylase lipase trypsinogen centroacinar cells intercalated duct start. Endocrine islet Langerhans alpha glucagon peripheral beta insulin central ~70% delta somatostatin PP epsilon ghrelin fenestrated capillaries no ducts.

Salivary: Serous acinus round nucleus basophilic basal eosinophilic apical zymogen watery protein. Mucous acinus flattened basal nucleus pale foamy mucin crescent serous demilune artifact myoepithelial. Mixed. Ducts intercalated cuboidal low myoepithelial → striated columnar basal striations mitochondria ion reabsorption Na+ water secretion K+ HCO_3^- → excretory stratified cuboidal→columnar. Myoepithelial around acini intercalated ducts squeezes.

Why liver fenestrated discontinuous sinusoids? Needs exchange proteins albumin clotting factors cells high permeability. Why bile canaliculi between hepatocytes sealed tight junctions? Bile toxic must be separated blood tight junctions blood-bile barrier.

Why pancreas acinar basophilic basal eosinophilic apical? Basal RER synthesizes enzymes apical stores zymogen polarity reflects flow like other protein-secreting cells.

Why striated duct basal striations? Mitochondria ion transport modifies saliva.

Seeing and Distinguishing

Recognition Logic

Look for... plates hepatocytes + sinusoids + central vein + portal triad liver. Basophilic basal + eosinophilic apical + centroacinar pancreatic serous acinar. Round clusters pale fenestrated capillaries no ducts islet. Striated basal mitochondria ducts salivary striated.

Then confirm... Kupffer macrophages sinusoids Ito vitamin A bile canaliculi between hepatocytes portal triad 3 structures.

Do not confuse with... liver vs pancreas both protein-secreting but liver plates + sinusoids + no zymogen granules polarized differently pancreas acinar + centroacinar + islet. Serous vs mucous serous basophilic basal eosinophilic apical round nucleus mucous pale foamy flat basal.

Decisive: central vein + portal triad + sinusoids discontinuous liver basophilic basal eosinophilic apical centroacinar pancreas exocrine pale cluster no duct central beta peripheral alpha islet striated duct basal striations salivary.

Compare & Distinguish

Clinical Meaning

Feature	Liver classic lobule	Portal lobule	Acinus
Center	Central vein	Portal triad	Zone 1 periportal to zone 3 perivenular
Flow	Blood to center	Bile to center	Blood portal→central metabolic gradient
Use	Structural	Bile drainage	Metabolic injury pattern
Feature	Serous acinus	Mucous acinus	
---	---	---	
Nucleus	Round central	Flat basal	
Cytoplasm	Basophilic basal eosinophilic apical	Pale foamy	
Secretion	Protein watery	Mucin viscous	
Location	Parotid pancreas	Sublingual	
Islet cell	Position	Hormone	
---	---	---	
Alpha	Peripheral	Glucagon raises glucose	
Beta	Central ~70%	Insulin lowers glucose	
Delta	Interspersed	Somatostatin inhibits	

Cirrhosis: Ito activation → collagen I fibrosis → disrupts sinusoid fenestrations → portal hypertension loss exchange.

Diabetes: beta loss type 1 autoimmune type 2 resistance.

Sjögren: autoimmune salivary + lacrimal → dry mouth dry eyes lymphocytic infiltration.

What to Remember

High-Yield

Must Know: Liver plates hepatocytes + sinusoids fenestrated discontinuous Kupffer macrophage Ito stellate vitamin A space Disse microvilli bile canaliculi tight junctions portal triad portal vein hepatic artery bile duct classic lobule central vein portal lobule portal center bile acinus zones 1 periportal oxygen rich 3 perivenular oxygen poor centrilobular necrosis. Gallbladder simple columnar no muscularis mucosae. Pancreas serous acinar basophilic basal eosinophilic apical centroacinar zymogen islet alpha peripheral glucagon beta central insulin delta somatostatin fenestrated no ducts. Salivary serous basophilic basal eosinophilic apical vs mucous pale foamy flat basal intercalated cuboidal myoepithelial → striated columnar basal striations mitochondria ion modification → excretory stratified.

Must Distinguish: Classic vs portal vs acinus serous vs mucous alpha vs beta position liver vs pancreas.

Must Understand: Why sinusoids discontinuous protein exchange why bile canaliculi tight junctions blood-bile barrier why striated duct modifies ions.

Common Misconceptions

Misconception: Liver lobule only one model. Reality three models explain different functions classic blood flow portal bile flow acinus metabolic zones injury.

Misconception: Pancreas islet has ducts. Reality endocrine no ducts fenestrated capillaries secrete blood.

Misconception: Serous demilune real. Reality often fixation artifact myoepithelial contraction.

Integrated Summary

Liver epithelial cords + sinusoids discontinuous fenestrated Kupffer Ito space Disse bile canaliculi tight junctions portal triad three models classic central vein blood portal triad bile acinus zones metabolic injury. Pancreas exocrine serous acinar basophilic basal eosinophilic apical centroacinar + endocrine islet alpha peripheral glucagon beta central insulin delta somatostatin no ducts fenestrated. Salivary serous protein vs mucous mucin mixed + duct system intercalated→striated basal mitochondria ion reabsorption→excretory + myoepithelial squeeze.

Check Your Understanding

1. Explain three liver models when each used.
2. Why liver needs discontinuous sinusoids and tight junction bile canaliculi?
3. Compare pancreatic acinar vs islet structure secretion ducts vascularity.
4. Distinguish serous vs mucous acinus and striated duct function.

Bridge

The gut absorbs nutrients, the liver processes them, and the pancreas regulates glucose. Next, an organ that exchanges gas with a similar barrier logic but for air: the respiratory system.

Rapid Review

- Liver plates hepatocytes eosinophilic + sinusoids fenestrated discontinuous Kupffer Ito space Disse microvilli bile canaliculi tight junctions portal triad classic central vein portal triad center acinus zones 1 periportal 3 perivenular centrilobular necrosis ischemia.
- Pancreas serous acinar basophilic basal eosinophilic apical centroacinar islet alpha peripheral glucagon beta central insulin delta somatostatin no ducts fenestrated.
- Salivary serous round nucleus basophilic basal eosinophilic apical protein mucous flat basal pale foamy mucin intercalated→striated basal mitochondria ion→excretory myoepithelial squeeze.

Chapter 14 — The Respiratory System

Junqueira 17th Edition — Chapter 17

> A bronchus has cartilage plates and submucosal glands, while a bronchiole has no cartilage or glands but has club cells and relatively more smooth muscle. The blood-air barrier is thin for diffusion yet tight to prevent protein leak.

Opening Question

How does an organ that must stay open also clean itself — and how does a 0.5 µm barrier let oxygen through but keep blood in?

Why This Matters

Asthma smooth muscle hyperplasia goblet hyperplasia emphysema alveolar wall loss hyaline membrane disease surfactant deficiency type II ciliary dyskinesia Kartagener.

Learning Objectives

By the end of this chapter, you will be able to:

1. Describe conducting vs respiratory portions and why conducting needs cartilage glands while respiratory needs thin barrier.
2. Distinguish bronchus vs bronchiole by cartilage glands epithelium smooth muscle Clara/club cells goblet.
3. Explain blood-air barrier 3 layers attenuated type I + fused basal laminae + endothelial and why ~0.5 µm.
4. Compare type I pneumocyte flat gas exchange ~95% surface vs type II cuboidal surfactant + progenitor.

The Landscape

Conducting portion conditions air cleans via mucociliary escalator pseudostratified ciliated + goblet + glands + cartilage. Respiratory portion exchanges gas via thin blood-air barrier type I + basal lamina + endothelium.

The Core Principle

Conducting = cartilage + glands + pseudostratified ciliated + goblet + smooth muscle; respiratory = no cartilage no glands simple cuboidal→squamous Clara/club alveoli type I flat + type II surfactant blood-air barrier 3 layers.

> CORE IDEA: A bronchus has cartilage plates and submucosal glands, while a bronchiole has no cartilage or glands but has club cells and relatively more smooth muscle. The blood-air barrier is thin for diffusion yet tight to prevent protein leak.

Building the System

Nasal cavity: respiratory region pseudostratified ciliated columnar + goblet + seromucous glands olfactory region pseudostratified columnar olfactory neurons + sustentacular + basal stem + Bowman glands.

Pharynx/larynx: stratified squamous non-keratinized where abrasion.

Trachea: C-shaped hyaline cartilage open posteriorly trachealis smooth muscle pseudostratified ciliated columnar + goblet + seromucous glands submucosa lamina propria elastic.

Bronchus: cartilage plates not C pseudostratified ciliated + goblet + glands smooth muscle spirals branching.

Bronchiole: no cartilage no glands epithelium simple columnar→cuboidal ciliated + Clara/club cells non-ciliated dome secretory surfactant-like progenitor detox cytochrome P450 smooth muscle prominent asthma goblet decreases.

Terminal bronchiole: last conducting simple cuboidal ciliated + Clara.

Respiratory bronchiole: conducting + alveoli outpocketings.

Alveolar duct: alveoli openings smooth muscle knobs.

Alveolar sac: clusters alveoli.

Alveolus: type I pneumocyte flat ~0.1 μm attenuated ~95% surface gas exchange tight junctions type II cuboidal lamellar bodies surfactant dipalmitoyl phosphatidylcholine reduces surface tension progenitor type I divides dust cell alveolar macrophage phagocytosis heart failure cells hemosiderin pores Kohn interalveolar.

Blood-air barrier: ~0.5 μm attenuated type I + fused basal lamina type IV collagen+laminin epithelial+endothelial + attenuated endothelial continuous non-fenestrated. Thin diffusion tight junctions prevent protein leak.

Why trachea C-shaped cartilage not O? Open posterior allows esophagus expansion swallowing trachealis muscle constricts.

Why bronchiole more smooth muscle proportionally? Bronchiole no cartilage muscle regulates airflow resistance asthma hyperreactivity.

Why blood-air barrier 3 layers fused basal lamina? Thin minimizes diffusion distance ~0.5 μm but fused basal lamina provides strength filtration prevents protein leak.

Why surfactant needed? Alveolus small radius high surface tension collapse surfactant reduces tension type II produces.

Seeing and Distinguishing

Recognition Logic

Look for... cartilage plates + glands + pseudostratified ciliated + goblet bronchus. No cartilage no glands + Clara dome non-ciliated + smooth muscle bronchiole. Single layer flat cells + capillaries + thin barrier + surfactant cells alveolus.

Then confirm... goblet present bronchus Clara present bronchiole lamellar bodies type II attenuated flat type I dust cells macrophages.

Do not confuse with... bronchus vs bronchiole bronchus cartilage + glands bronchiole none + Clara. Type I vs endothelial both flat but type I epithelium tight junctions + surfactant type II nearby endothelial continuous.

Decisive: cartilage+glands bronchus no cartilage no glands + Clara bronchiole flat attenuated + fused basal lamina + type II lamellar alveolus blood-air barrier.

Compare & Distinguish

Clinical Meaning

Feature	Bronchus	Bronchiole	Alveolus
Cartilage	Plates	None	None
Glands	Yes submucosal	None	None
Epithelium	Pseudostratified ciliated + goblet	Simple columnar→cuboidal ciliated + Clara no goblet	Type I flat + Type II cuboidal
Smooth muscle	Spirals	Prominent relative	Knobs duct
Clara/club	No	Yes	No
Feature	Type I pneumocyte	Type II pneumocyte	
---	---	---	
Shape	Flat attenuated ~0.1 μm	Cuboidal	
Surface	~95%	~5%	
Organelle	Few tight junctions	Lamellar bodies surfactant RER	
Function	Gas exchange	Surfactant + progenitor	
Division	No	Yes progenitor type I	

Asthma: smooth muscle hyperplasia/hypertrophy bronchiole goblet hyperplasia basement membrane thickening eosinophils.

Emphysema: alveolar wall destruction loss surface decreased recoil smoking.

Hyaline membrane disease RDS: premature type II immature surfactant deficiency → alveolar collapse hyaline membranes.

What to Remember

High-Yield

Must Know: Conducting vs respiratory trachea C-shaped hyaline cartilage pseudostratified ciliated goblet glands bronchus plates cartilage glands pseudostratified goblet smooth muscle bronchiole no cartilage no glands simple columnar/cuboidal ciliated + Clara dome secretory progenitor smooth muscle prominent terminal bronchiole last conducting respiratory bronchiole alveoli outpocket alveolar duct sac alveolus type I flat attenuated ~95% gas exchange tight junctions type II cuboidal lamellar surfactant progenitor dust cell macrophage blood-air barrier 3 layers attenuated type I + fused basal lamina type IV+laminin + attenuated endothelial ~0.5 µm.

Must Distinguish: Bronchus vs bronchiole vs alveolus type I vs II continuous vs fenestrated capillary alveolar continuous.

Must Understand: Why C-shaped why smooth muscle more bronchiole why 3-layer barrier thin but tight why surfactant prevents collapse.

Common Misconceptions

Misconception: Alveolar capillary fenestrated. Reality continuous non-fenestrated attenuated tight junctions fused basal lamina.

Misconception: Goblet cells bronchioles many. Reality decreases distally Clara increases.

Misconception: Type I divides. Reality Type II progenitor.

Integrated Summary

Respiratory conducting conditions cleans nasal pseudostratified ciliated goblet trachea C-shaped hyaline cartilage pseudostratified ciliated goblet glands bronchus plates cartilage glands pseudostratified goblet bronchiole no cartilage no glands simple ciliated + Clara progenitor smooth muscle. Respiratory exchanges respiratory bronchiole alveoli alveolar duct sac alveolus type I flat 95% gas exchange + type II cuboidal lamellar surfactant progenitor + dust cell macrophage blood-air barrier attenuated type I + fused basal lamina + attenuated endothelial ~0.5 µm thin diffusion tight protein.

Check Your Understanding

1. Distinguish bronchus vs bronchiole 5 decisive features.
2. Explain blood-air barrier layers why ~0.5 µm.
3. Compare type I vs II shape surface organelle function division.
4. Why trachea C-shaped cartilage?
5. Why asthma affects bronchiole smooth muscle goblet?

Bridge

The lung exchanges gas with a thin barrier. Skin protects with a thick barrier. Next, a stratified squamous keratinized specialization: skin.

Rapid Review

- Conducting nasal pseudostratified ciliated goblet trachea C hyaline cartilage pseudostratified ciliated goblet glands bronchus plates cartilage glands pseudostratified goblet smooth muscle bronchiole no cartilage no glands simple ciliated + Clara dome smooth muscle prominent.
- Respiratory respiratory bronchiole alveoli alveolar duct sac alveolus type I flat attenuated 95% gas exchange type II cuboidal lamellar surfactant progenitor dust cell macrophage.
- Barrier type I + fused basal lamina type IV+laminin + endothelial ~0.5 μm continuous non-fenestrated.
- Asthma smooth muscle hyperplasia goblet hyperplasia emphysema wall loss RDS surfactant deficiency type II.

Chapter 15 — Skin

Junqueira 17th Edition — Chapter 18

> In skin, basal stem cells attach via hemidesmosomes, spinous cells connect via desmosomes visible as spines, granular cells contain keratohyalin, and corneum forms a keratin barrier. Melanocytes provide a supranuclear melanin cap for UV protection, Langerhans cells present antigen, and Merkel cells mediate touch.

Opening Question

How does skin be waterproof, touch-sensitive, and able to regrow after a cut?

Why This Matters

Basal cell carcinoma basal layer melanoma melanocyte Langerhans antigen presentation burns stem cell survival determines healing Meissner touch Pacinian pressure.

Learning Objectives

By the end of this chapter, you will be able to:

1. Describe epidermal layers basale→spinosum→granulosum→lucidum→corneum and keratinization process.
2. Identify melanocyte Langerhans Merkel keratinocyte by function location.
3. Distinguish papillary loose Meissner vs reticular dense irregular Pacinian dermis.
4. Compare thick vs thin skin and eccrine vs apocrine sweat glands and sebaceous holocrine.

The Landscape

Skin = keratinized stratified squamous epithelium epidermis + CT dermis papillary loose + reticular dense irregular + appendages. Keratinization terminal differentiation.

The Core Principle

Epidermis basal stem → spinous desmosomes → granular keratohyalin → corneum anucleate keratin waterproof; dermis papillary loose Meissner touch reticular dense irregular Pacinian pressure; appendages hair + arrector pili smooth muscle + sebaceous holocrine + eccrine merocrine.

> CORE IDEA: In skin, basal stem cells attach via hemidesmosomes, spinous cells connect via desmosomes visible as spines, granular cells contain keratohyalin, and corneum forms a keratin barrier. Melanocytes provide a supranuclear melanin cap for UV protection, Langerhans cells present antigen, and Merkel cells mediate touch.

Building the System

Epidermis:

- Basale single layer cuboidal stem hemidesmosomes integrin laminin mitosis melanocytes neural crest dendritic melanosomes melanin supranuclear cap UV protection DOPA.
- Spinosum several layers polyhedral desmosomes intercellular bridges spines keratin intermediate filaments Langerhans cells dendritic Birbeck granules tennis racket antigen presentation monocyte.
- Granulosum 3-5 layers flattened keratohyalin granules profilaggrin + lamellar bodies lipids waterproof.
- Lucidum thick skin only translucent anucleate eleidin.
- Corneum many layers anucleate flat keratin squames waterproof desquamation.

Other cells: Melanocyte dendritic basal melanosomes melanin transferred keratinocytes supranuclear cap. Langerhans dendritic suprabasal antigen presentation Birbeck. Merkel basal touch associated sensory nerve dense granules.

Dermis: Papillary loose CT dermal papillae Meissner corpuscle touch stacked Schwann capillaries. Reticular dense irregular CT collagen I + elastic Pacinian corpuscle pressure onion layers hair follicles glands.

Appendages: Hair follicle bulb matrix stem papilla CT arrector pili smooth muscle sympathetic goose bumps hair. Sebaceous holocrine whole cell disintegrates lipid sebum associated hair except lips androgen regulated. Eccrine sweat merocrine simple coiled tubular clear+dark cells duct stratified cuboidal thermoregulation palms/soles many. Apocrine merocrine despite name coiled tubular myoepithelial axilla/perineal odor puberty lipid.

Thick vs thin: Thick palms/soles thick epidermis with lucidum thick corneum no hair no sebaceous many eccrine Meissner many. Thin rest thin epidermis no lucidum hair + sebaceous + eccrine + apocrine.

Why basale hemidesmosomes and spinosum desmosomes? Basale needs anchorage basement membrane hemidesmosome integrin laminin spinosum needs cell-cell adhesion resist shear desmosome desmoglein keratin.

Why corneum anucleate keratin waterproof? Keratin intermediate filaments + lipid lamellar bodies fill intercellular space barrier.

Why melanin supranuclear cap? Protects nucleus DNA UV.

Seeing and Distinguishing

Recognition Logic

Look for... keratinized stratified squamous + dermis + appendages skin. Thick corneum + lucidum + no hair thick skin. Thin corneum + hair +

sebaceous thin skin.

Then confirm... basal cuboidal stem + spinosum desmosome spines + granulosum keratohyalin + corneum anucleate melanocyte dendritic basal Langerhans dendritic suprabasal Merkel basal touch Meissner papillary touch Pacinian reticular pressure.

Do not confuse with... thick vs thin skin papillary vs reticular dermis eccrine vs apocrine vs sebaceous holocrine vs merocrine.

Decisive: lucidum + no hair + very thick corneum thick skin hair + sebaceous thin Meissner papillary Pacinian reticular onion.

Compare & Distinguish

Clinical Meaning

Feature	Thick skin	Thin skin	
Location	Palms soles	Rest	
Epidermis	Thick + lucidum	Thin no lucidum	
Corneum	Very thick	Thin	
Hair/sebaceous	No	Yes	
Eccrine	Many	Fewer	
Meissner	Many	Fewer	
Feature	Papillary dermis	Reticular dermis	
---	---	---	
CT	Loose	Dense irregular	
Corpuscle	Meissner touch	Pacinian pressure	
Location	Superficial	Deep	
Gland	Secretion	Mechanism	Location
Sebaceous	Lipid sebum	Holocrine whole cell	Hair associated
Eccrine	Watery sweat	Merocrine exocytosis	Palms soles many
Apocrine	Viscous odor	Merocrine	Axilla perineal puberty

Basal cell carcinoma: basal layer most common skin cancer locally invasive.

Melanoma: melanocyte aggressive ABCDE.

Burns: superficial heals basal stem appendages deep needs graft.

What to Remember

High-Yield

Must Know: Layers basale cuboidal stem hemidesmosome melanocyte + spinosum polyhedral desmosome spines Langerhans + granulosum keratohyalin lamellar bodies + lucidum thick only + corneum anucleate keratin waterproof melanocyte dendritic basal melanosome supranuclear cap DOPA Langerhans dendritic suprabasal Birbeck antigen Merkel basal touch papillary loose Meissner touch reticular dense irregular Pacinian pressure hair bulb matrix arrector pili smooth muscle sebaceous holocrine lipid eccrine merocrine watery thermoregulation apocrine merocrine viscous odor axilla.

Must Distinguish: Thick vs thin papillary vs reticular sebaceous holocrine vs eccrine/apocrine merocrine Meissner vs Pacinian melanocyte vs Langerhans vs Merkel.

Must Understand: Why hemidesmosome basal vs desmosome spinosum why corneum waterproof keratin+lipid why melanin supranuclear cap UV protection.

Common Misconceptions

Misconception: Sebaceous merocrine. Reality holocrine whole cell disintegrates.

Misconception: Apocrine apocrine mechanism. Reality despite name merocrine exocytosis lipid component.

Misconception: Thick skin has hair. Reality no hair no sebaceous.

Integrated Summary

Skin keratinized stratified squamous epidermis basal stem hemidesmosome melanocyte dendritic melanosome supranuclear cap + spinosum desmosome spines Langerhans Birbeck + granulosum keratohyalin lamellar bodies + lucidum thick only + corneum anucleate keratin waterproof + Merkel touch dermis papillary loose Meissner touch + reticular dense irregular Pacinian pressure appendages hair bulb matrix arrector pili smooth muscle + sebaceous holocrine + eccrine merocrine watery + apocrine merocrine viscous.

Check Your Understanding

1. Reconstruct epidermal layers events each.
2. Distinguish melanocyte vs Langerhans vs Merkel location function.
3. Compare thick vs thin skin papillary vs reticular dermis.
4. Why superficial burn heals from appendages while deep needs graft?
5. Compare sebaceous vs eccrine vs apocrine mechanism location.

Bridge

Skin is the outer barrier. Next, ductless glands that command long-distance signaling via blood: endocrine glands.

Rapid Review

- Layers basale cuboidal stem hemidesmosome + spinosum desmosome spines + granulosum keratohyalin lamellar + lucidum thick only + corneum anucleate keratin.
- Cells melanocyte dendritic basal melanosome supranuclear cap Langerhans dendritic suprabasal Birbeck antigen Merkel basal touch keratinocyte keratin.
- Dermis papillary loose Meissner touch reticular dense irregular Pacinian pressure.
- Appendages hair bulb matrix arrector pili smooth sebaceous holocrine lipid eccrine merocrine watery apocrine merocrine viscous axilla.
- Thick palms soles thick corneum lucidum no hair sebaceous many eccrine Meissner thin rest hair sebaceous.

Chapter 16 — Endocrine Glands

Junqueira 17th Edition — Chapter 20

> In endocrine glands, anterior pituitary acidophils produce growth hormone and prolactin, basophils produce FSH, LH, ACTH, and TSH, with modern immunohistochemistry identifying somatotrophs, lactotrophs, corticotrophs, gonadotrophs, and thyrotrophs. Posterior pituitary stores ADH and oxytocin in Herring bodies. Thyroid follicles store colloid for T3 and T4 with parafollicular C cells producing calcitonin. Adrenal cortex shows outer to inner zonation for salt, sugar, and sex steroids, and medulla contains chromaffin cells producing catecholamines.

Opening Question

How does a pea-sized gland behind eyes control growth, milk, stress, thirst — with only handful of cell types?

Why This Matters

Prolactinoma most common pituitary adenoma bitemporal hemianopia optic chiasm compression Graves scalloped colloid Cushing fasciculata cortisol pheochromocytoma medulla catecholamines diabetes insipidus ADH storage failure.

Learning Objectives

By the end of this chapter, you will be able to:

1. Describe pituitary anterior vs posterior origin cell types hormones why posterior stores not synthesizes.
2. Recite anterior pituitary cell-hormone pairing modern IHC somatotroph GH lactotroph prolactin corticotroph ACTH gonadotroph FSH/LH thyrotroph TSH and historical tinctorial acidophil/basophil/chromophobe.
3. Describe thyroid follicle colloid thyroglobulin follicular cells T3/T4 + parafollicular C cells calcitonin.
4. Describe parathyroid chief PTH + oxyphil.
5. Describe adrenal cortex zones glomerulosa aldosterone salt fasciculata cortisol sugar clear cells SER + tubular mitochondria reticularis androgen sex and medulla chromaffin catecholamines modified postganglionic sympathetic.
6. Identify pineal pinealocytes + corpora arenacea brain sand.

The Landscape

Endocrine = ductless secretion into blood fenestrated capillaries. Identified by signature cell/hormone pairing. Pituitary divides system anterior glandular synthesizes posterior neural stores hypothalamic hormones.

The Core Principle

Gland = cell + hormone pituitary anterior synthesizes via portal system posterior stores via axonal transport adrenal cortex outer→inner salt→sugar→sex.

> CORE IDEA: In endocrine glands, anterior pituitary acidophils produce growth hormone and prolactin, basophils produce FSH, LH, ACTH, and TSH, with modern immunohistochemistry identifying somatotrophs, lactotrophs, corticotrophs, gonadotrophs, and thyrotrophs. Posterior pituitary stores ADH and oxytocin in Herring bodies. Thyroid follicles store colloid for T3 and T4 with parafollicular C cells producing calcitonin. Adrenal cortex shows outer to inner zonation for salt, sugar, and sex steroids, and medulla contains chromaffin cells producing catecholamines.

Building the System

Pituitary hypophysis: sella turcica dual origin.

- Anterior adenohypophysis oral ectoderm Rathke pouch glandular synthesizes.
- Cells somatotroph GH acidophil lactotroph prolactin acidophil corticotroph ACTH basophil gonadotroph FSH/LH basophil thyrotroph TSH basophil chromophobes degranulated stem.
- Historical tinctorial acidophils eosinophilic GH prolactin basophils basophilic B-FLAT chromophobes pale. Modern IHC hormone-specific more accurate.
- Posterior neurohypophysis neural ectoderm neural stores hypothalamic hormones Herring bodies dilated axon terminals neurosecretory granules ADH/oxytocin pituicytes glia.
- Intermediate pars intermedia.
- Portal system hypothalamus releasing hormones → primary plexus → portal veins → secondary plexus anterior → stimulates.

Thyroid: follicles colloid thyroglobulin iodinated follicular cells simple cuboidal→columnar T3/T4 synthesis RER basal basophilic apical eosinophilic parafollicular C cells calcitonin clear neural crest fenestrated capillaries.

Parathyroid: chief cells small clear central nucleus PTH + oxyphil large eosinophilic mitochondria-rich older cords fat increases age.

Adrenal: cortex mesoderm + medulla neural crest.

- Cortex zones outer→inner GFR glomerulosa arched aldosterone salt mineralocorticoid fasciculata cords clear cells foamy SER + tubular mitochondria lipid droplets cortisol sugar glucocorticoid reticularis anastomosing androgen sex lipofuscin.
- Medulla chromaffin cells modified postganglionic sympathetic no axon catecholamines epinephrine/norepinephrine chromaffin reaction brown sympathetic ganglion.

Pineal: pinealocytes melatonin corpora arenacea brain sand calcification glia.

Why posterior stores not synthesizes? ADH/oxytocin synthesized hypothalamus supraoptic/paraventricular cell bodies transported via axons posterior Herring bodies stored released. So lesion hypothalamus affects posterior.

Why adrenal cortex clear cells fasciculata? Steroid synthesis needs SER and tubular-cristae mitochondria and lipid droplets cholesterol clear foamy.

Why thyroid colloid? Thyroglobulin storage in colloid is distinctive among endocrine glands because hormone precursor is stored extracellularly, whereas most endocrine glands store hormone intracellularly.

Seeing and Distinguishing

Recognition Logic

Look for... acidophils eosinophilic + basophils basophilic + chromophobes pale + Herring bodies pale pituitary. Colloid follicles + cuboidal cells + C cells clear thyroid. Chief small clear + oxyphil large eosinophilic parathyroid. GFR zones glomerulosa arched fasciculata clear cords reticularis anastomosing + medulla chromaffin adrenal. Brain sand pineal.

Then confirm... colloid scalloped active thyroid Graves clear cells foamy fasciculata cortisol Herring bodies posterior stores.

Do not confuse with... thyroid vs parathyroid thyroid colloid follicles parathyroid no colloid chief cords. Adrenal cortex vs medulla cortex clear foamy steroid medulla chromaffin catecholamines. Anterior vs posterior pituitary anterior glandular acidophil/basophil posterior neural Herring bodies.

Decisive: colloid thyroid no colloid chief cords parathyroid GFR salt sugar sex adrenal cortex chromaffin medulla Herring bodies posterior pituitary brain sand pineal.

Compare & Distinguish

Clinical Meaning

Gland	Cell	Hormone	Stain/feature
Anterior acidophil somatotroph	GH	Growth	Eosinophilic
Anterior acidophil lactotroph	Prolactin	Milk	Eosinophilic
Anterior basophil corticotroph	ACTH	Stress	Basophilic B-FLAT
Anterior basophil gonadotroph	FSH/LH	Gonads	Basophilic B-FLAT

Gland	Cell	Hormone	Stain/feature
Anterior basophil thyrotroph	TSH	Thyroid	Basophilic B-FLAT
Posterior	Herring bodies	ADH/oxytocin stored	Pale neurosecretory
Thyroid follicular	T3/T4	Metabolism	Colloid cuboidal
Thyroid C	Calcitonin	Calcium lower	Clear parafollicular
Parathyroid chief	PTH	Calcium raise	Small clear
Parathyroid oxyphil	—	—	Large eosinophilic
Adrenal glomerulosa	Aldosterone	Salt	Arched outer
Adrenal fasciculata	Cortisol	Sugar	Clear foamy cords
Adrenal reticularis	Androgen	Sex	Anastomosing lipofuscin
Adrenal medulla chromaffin	Catecholamines	Stress	Modified sympathetic

Prolactinoma: most common pituitary adenoma lactotroph acidophil prolactin galactorrhea amenorrhea bitemporal hemianopia optic chiasm compression above sella.

Graves: TSH receptor antibody → follicular hyperplasia scalloped colloid active.

Hashimoto: lymphocytic infiltration thyroid.

What to Remember

High-Yield

Must Know: Pituitary anterior adenohypophysis oral ectoderm acidophils GH prolactin basophils B-FLAT FSH LH ACTH TSH chromophobes posterior neurohypophysis neural ectoderm Herring bodies ADH oxytocin stored pituicytes portal system hypothalamus releasing hormones. Thyroid follicles colloid thyroglobulin follicular T3/T4 + C calcitonin clear. Parathyroid chief PTH small clear + oxyphil large eosinophilic. Adrenal cortex GFR glomerulosa aldosterone salt arched fasciculata cortisol sugar clear foamy SER tubular mitochondria reticularis androgen sex lipofuscin anastomosing medulla chromaffin catecholamines modified postganglionic sympathetic. Pineal pinealocyte melatonin corpora arenacea brain sand.

Must Distinguish: Anterior vs posterior pituitary thyroid vs parathyroid cortex zones GFR cortex vs medulla acidophil vs basophil historical vs modern IHC.

Must Understand: Why posterior stores not synthesizes why clear cells fasciculata steroid why colloid storage extracellular.

Common Misconceptions

Misconception: Posterior pituitary synthesizes ADH. Reality synthesized hypothalamus stored posterior Herring bodies most tested pituitary fact.

Misconception: Acidophil/basophil modern classification. Reality historical tinctorial modern IHC somatotroph lactotroph corticotroph gonadotroph thyrotroph more accurate but B-FLAT mnemonic still useful.

Misconception: Thyroid C cells follicular origin. Reality neural crest parafollicular calcitonin.

Integrated Summary

Endocrine ductless fenestrated capillaries cell+ hormone pairing pituitary dual origin anterior glandular synthesizes acidophils GH prolactin basophils B-FLAT FSH LH ACTH TSH modern somatotroph lactotroph corticotroph gonadotroph thyrotroph + posterior neural stores ADH oxytocin Herring bodies portal system thyroid colloid follicles T3/T4 + C calcitonin parathyroid chief PTH + oxyphil adrenal cortex GFR glomerulosa aldosterone salt fasciculata cortisol sugar clear cells reticularis androgen sex + medulla chromaffin catecholamines modified sympathetic pineal pinealocyte melatonin brain sand.

Check Your Understanding

1. Explain why posterior pituitary stores not synthesizes where hormones made.
2. Reconstruct anterior pituitary cell-hormone pairing both historical acidophil/basophil and modern IHC.
3. Distinguish thyroid vs parathyroid decisive features.
4. Explain adrenal cortex GFR salt sugar sex outer to inner.
5. Why prolactinoma associated bitemporal hemianopia?

Bridge

Endocrine glands complete the body's long-distance signaling. You now have a continuous map from methods to cells to tissues to organ systems to hormonal control. The sixteen chapters read as one connected course.

Rapid Review

- Pituitary anterior adenohypophysis oral ectoderm acidophils GH PRL basophils B-FLAT FSH LH ACTH TSH chromophobes posterior neurohypophysis neural Herring bodies ADH oxytocin stored pituicytes portal hypothalamus.
- Thyroid colloid follicles T3/T4 + C calcitonin clear parafollicular neural crest.
- Parathyroid chief PTH small clear + oxyphil large eosinophilic.
- Adrenal cortex GFR glomerulosa aldosterone salt arched fasciculata cortisol sugar clear foamy SER tubular mitochondria reticularis androgen sex lipofuscin medulla chromaffin catecholamines modified

sympathetic.

- Pineal pinealocyte melatonin corpora arenacea brain sand.

- Prolactinoma most common bitemporal hemianopia Graves scalloped colloid Cushing fasciculata pheochromocytoma medulla DI ADH.